

Human review packet: LBG24SpatialUnderreporting

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: LBG24SpatialUnderreporting

Paper id: LBG24SpatialUnderreporting

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Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper LBG24SpatialUnderreporting --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

Contents

- [Governing Lean declarations \(65; supporting context\)](#)
- [Semantic library prerequisites \(2; not paper claims\)](#)
 - [AppliedModelingLib.Probability.PoissonProcess.rateExposureParam](#)
 - [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [Paper-specific semantic prerequisites \(14; not paper claims\)](#)
 - [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
 - [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
 - [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
 - [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.iidGapLaw](#)
 - [LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition](#)
 - [LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel](#)
 - [LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition](#)
 - [LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel](#)
 - [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel](#)
 - [LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation](#)
 - [LBG24SpatialUnderreporting.equation33_nyc_observation_end](#)
 - [LBG24SpatialUnderreporting.equation34_chicago_observation_end](#)
 - [LBG24SpatialUnderreporting.poissonRegressionRate](#)
 - [LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood](#)
- [Source claims \(10\)](#)
 - [sourceHomogeneousReportingDelayMeanSpec](#)
 - [sourceTheorem1LikelihoodDecompositionSpec](#)
 - [sourceEquation2PoissonCountPMFSpec](#)
 - [sourceEquation3MaximumLikelihoodEstimateSpec](#)
 - [sourceEquation6PoissonRegressionLikelihoodSpec](#)
 - [sourceZeroInflatedLikelihoodExtensionSpec](#)
 - [sourceLemma1CalendarTimeDurationObservedProcessSpec](#)
 - [sourceProposition1CalendarTimeNonidentifiabilitySpec](#)
 - [sourceLemma2SelectedStartExponentialTailSpec](#)
 - [sourcePostFirstJumpPoissonShiftSpec](#)

Governing Lean declarations

These exact graph-bound declaration bodies are retained by the displayed specifications or reviewed prerequisites. They are supporting context and do not add source-result rows or semantic judgments.

AppliedModelingLib.Probability.Exponential.Model

Declaration location: reusable library

Lean declaration body

field rate:
AppliedModelingLib.Probability.Exponential.Model $\rightarrow \mathbb{R}$

field rate_pos:
 $\forall$ (self : AppliedModelingLib.Probability.Exponential.Model), $0 < \text{self.rate}$

AppliedModelingLib.Probability.Exponential.Model.measure

Declaration location: reusable library

Lean declaration body

type:
AppliedModelingLib.Probability.Exponential.Model $\rightarrow \text{MeasureTheory.Measure } \mathbb{R}$

value:
fun (M : AppliedModelingLib.Probability.Exponential.Model) => ProbabilityTheory.expMeasure M.rate

Direct retained dependencies

- [AppliedModelingLib.Probability.Exponential.Model](#)

AppliedModelingLib.Probability.PoissonCountingMeasureByLaw

Declaration location: reusable library

Lean declaration body

field isProbability:
 $\forall \{ \Omega : \text{Type } u_1 \} \{ X : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{MeasurableSpace } X] \{ P : \text{MeasureTheory.Measure } \Omega \}$
 {intensity : MeasureTheory.Measure X}
 (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity),
 MeasureTheory.IsProbabilityMeasure P

field finiteCount:
{ $\Omega : \text{Type } u_1$ } $\rightarrow$
 { $X : \text{Type } u_2$ } $\rightarrow$
 [inst : MeasurableSpace Ω] $\rightarrow$
 [inst_1 : MeasurableSpace X] $\rightarrow$
 {P : MeasureTheory.Measure Ω } $\rightarrow$
 {intensity : MeasureTheory.Measure X} $\rightarrow$
 AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity $\rightarrow \text{Set } X \rightarrow \Omega \rightarrow \mathbb{N}$

field finiteCount_measurable:
 $\forall \{ \Omega : \text{Type } u_1 \} \{ X : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{MeasurableSpace } X] \{ P : \text{MeasureTheory.Measure } \Omega \}$
 {intensity : MeasureTheory.Measure X}
 (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s : Set X),
 Measurable (self.finiteCount s)

field finiteCount_empty_ae:
 $\forall \{ \Omega : \text{Type } u_1 \} \{ X : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{MeasurableSpace } X] \{ P : \text{MeasureTheory.Measure } \Omega \}$
 {intensity : MeasureTheory.Measure X}
 (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity),
 $\forall^m (\omega : \Omega) \partial P, \text{self.finiteCount } \emptyset \omega = 0$

field finiteCount_mono:
 $\forall \{ \Omega : \text{Type } u_1 \} \{ X : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{MeasurableSpace } X] \{ P : \text{MeasureTheory.Measure } \Omega \}$
 {intensity : MeasureTheory.Measure X}
 (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s t : Set X),
 MeasurableSet s $\rightarrow$
 MeasurableSet t $\rightarrow$ intensity s $\neq \top \rightarrow$ intensity t $\neq \top \rightarrow s \subseteq t \rightarrow \forall (\omega : \Omega), \text{self.finiteCount } s \omega \leq \text{self.finiteCount } t \omega$

field finiteCount_union_ae:
 $\forall \{ \Omega : \text{Type } u_1 \} \{ X : \text{Type } u_2 \} [\text{inst} : \text{MeasurableSpace } \Omega] [\text{inst}_1 : \text{MeasurableSpace } X] \{ P : \text{MeasureTheory.Measure } \Omega \}$
 {intensity : MeasureTheory.Measure X}
 (self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s t : Set X),
 MeasurableSet s $\rightarrow$
 MeasurableSet t $\rightarrow$
 intensity s $\neq \top \rightarrow$
 intensity t $\neq \top \rightarrow$
 Disjoint s t $\rightarrow \forall^m (\omega : \Omega) \partial P, \text{self.finiteCount } (s \cup t) \omega = \text{self.finiteCount } s \omega + \text{self.finiteCount } t \omega$

field finiteCount_independent:

```

∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (n : ℕ) (s : Fin n → Set X),
(∀ (i : Fin n), MeasurableSet (s i)) →
(∀ (i : Fin n), intensity (s i) ≠ 0) →
(Pairwise fun (i j : Fin n) => Disjoint (s i) (s j)) →
ProbabilityTheory.IndepFun (fun (i : Fin n) => self.finiteCount (s i)) P

field finiteCount_hasLaw:
∀ {Ω : Type u_1} {X : Type u_2} [inst : MeasurableSpace Ω] [inst_1 : MeasurableSpace X] {P : MeasureTheory.Measure Ω}
{intensity : MeasureTheory.Measure X}
(self : AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω X P intensity) (s : Set X),
MeasurableSet s →
intensity s ≠ 0 →
ProbabilityTheory.HasLaw (self.finiteCount s) (ProbabilityTheory.poissonMeasure (intensity s).toNNReal) P

```

AppliedModelingLib.Probability.PoissonProcess.ObservationWindow

Declaration location: reusable library

Lean declaration body

```

field startTime:
AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

field endTime:
AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

field start_le_end:
∀ (self : AppliedModelingLib.Probability.PoissonProcess.ObservationWindow), self.startTime ≤ self.endTime

```

AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure

Declaration location: reusable library

Lean declaration body

```

type:
AppliedModelingLib.Probability.PoissonProcess.ObservationWindow → ℝ

value:
fun (W : AppliedModelingLib.Probability.PoissonProcess.ObservationWindow) => W.endTime - W.startTime

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)

AppliedModelingLib.Probability.PoissonProcess.RateIndependent

Declaration location: reusable library

Lean declaration body

```

type:
(ℝ → ℝ) → Prop

value:
fun (f : ℝ → ℝ) => ∃ (c : ℝ), ∀ (rate : ℝ), f rate = c

```

AppliedModelingLib.Probability.PoissonProcess.countLikelihood

Declaration location: reusable library

Lean declaration body

```

type:
ℝ → ℝ → ℕ → ℝ

value:
fun (rate exposure : ℝ) (n : ℕ) => Real.exp (-(rate * exposure)) * (rate * exposure) ^ n / ↑n.factorial

```

AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector

Declaration location: reusable library

Lean declaration body

```
type:
(q : ℕ) → (Fin q → ℝ) → Fin q → ℝ

value:
fun (q : ℕ) (a : Fin q → ℝ) (a_1 : Fin q) => ∑ j : { j : Fin q // j ≤ a_1 }, a ↑j
```

AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → (q : ℕ) → (Fin q → ℝ) → ENNReal

value:
fun (rate : ℝ) (q : ℕ) (a : Fin q → ℝ) => ∏ i : Fin q, ProbabilityTheory.exponentialPDF rate (a i)
```

AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure

Declaration location: reusable library

Lean declaration body

```
type:
ℝ → MeasureTheory.Measure (ℕ → ℝ)

value:
fun (rate : ℝ) => MeasureTheory.Measure.infinitePi fun (x : ℕ) => ProbabilityTheory.expMeasure rate
```

AppliedModelingLib.Probability.PoissonProcess.interarrival

Declaration location: reusable library

Lean declaration body

```
type:
ℕ → (ℕ → ℝ) → ℝ

value:
fun (n : ℕ) (a : ℕ → ℝ) => a n
```

AppliedModelingLib.Probability.PoissonProcess.arrivalTime

Declaration location: reusable library

Lean declaration body

```
type:
ℕ → (ℕ → ℝ) → ℝ

value:
fun (n : ℕ) (a : ℕ → ℝ) => ∑ i ∈ Finset.range (n + 1), AppliedModelingLib.Probability.PoissonProcess.interarrival i a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.interarrival](#)

AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival

Declaration location: reusable library

Lean declaration body

```
type:
(ℕ → ℝ) → ℝ

value:
fun (a : ℕ → ℝ) => AppliedModelingLib.Probability.PoissonProcess.interarrival 0 a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.interarrival](#)

AppliedModelingLib.Probability.PoissonProcess.canonicalRenewalCount

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{N}$ 

value:
fun (t :  $\mathbb{R}$ ) (a :  $\mathbb{N} \rightarrow \mathbb{R}$ ) =>
  if h :  $\exists (n : \mathbb{N}), t < \text{AppliedModelingLib.Probability.PoissonProcess.arrivalTime } n \ a$  then Nat.find h else 0
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.arrivalTime](#)

AppliedModelingLib.Probability.PoissonProcess.futureInterarrival

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (n :  $\mathbb{N}$ ) (a :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (a_1 :  $\mathbb{N}$ ) => AppliedModelingLib.Probability.PoissonProcess.interarrival (n + a_1) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.interarrival](#)

AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (start :  $\mathbb{R}$ ) (jumpTime :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (a :  $\mathbb{N}$ ) =>
  match a with
  | 0 => start
  | i.succ => jumpTime i
```

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline

Declaration location: reusable library

Lean declaration body

```
field window:
  AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline →
    AppliedModelingLib.Probability.PoissonProcess.ObservationWindow

field count:
  AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline →  $\mathbb{N}$ 

field jumpTime:
  AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline →  $\mathbb{N} \rightarrow \mathbb{R}$ 

field endpoint_mono:
 $\forall (self : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}),$ 
  Monotone (AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint self.window.startTime self.jumpTime)

field last_le_end:
 $\forall (self : \text{AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}),$ 
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint self.window.startTime self.jumpTime self.count  $\leq$ 
    self.window.endTime
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (start :  $\mathbb{R}$ ) (jumpTime :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (i :  $\mathbb{N}$ ) =>
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime (i + 1) -
  AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime i
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap

Declaration location: reusable library

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → Fin T.count →  $\mathbb{R}$ 

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) (j : Fin T.count) =>
  AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes T.window.startTime T.jumpTime ↑j
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.interarrivalGapFromJumpTimes](#)

AppliedModelingLib.Probability.PoissonProcess.noArrivalProb

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (rate exposure :  $\mathbb{R}$ ) => Real.exp (-(rate * exposure))
```

AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 

value:
fun (count :  $\mathbb{N}$ ) (exposure rate :  $\mathbb{R}$ ) => ↑count * Real.log rate - rate * exposure
```

AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.intervalCount

Declaration location: reusable library

Lean declaration body

```
type:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P →
NNReal → NNReal → Ω → ℕ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
(H : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) (s t : NNReal)
(ω : Ω) =>
H.count t ω - H.count s ω
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)

AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw

Declaration location: reusable library

Lean declaration body

```
type:
{rate : ℝ} →
0 < rate →
AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw (ℕ → ℝ)
(AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate)

value:
fun {rate : ℝ} (hrate : 0 < rate) =>
{
isProbability :=
AppliedModelingLib.Probability.PoissonProcess.isProbabilityMeasure_exponentialInterarrivalMeasure hrate,
rate := rate, rate_pos := hrate,
count := fun (t : NNReal) (ω : ℕ → ℝ) => AppliedModelingLib.Probability.PoissonProcess.canonicalRenewalCount (↑t) ω,
count_measurable :=
AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw._proof_1,
count_zero ae :=
AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw._proof_3
hrate,
count_mono ae :=
AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw._proof_4
hrate,
hasIndepIncrements := AppliedModelingLib.Probability.PoissonProcess.canonicalForwardCount_hasIndepIncrements hrate,
increment_hasLaw :=
@AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw._proof_6
rate hrate }
}
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [AppliedModelingLib.Probability.PoissonProcess.canonicalRenewalCount](#)
- [AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure](#)
- [AppliedModelingLib.Probability.PoissonProcess.rateExposureParam](#)

AppliedModelingLib.Probability.PoissonProcess.forwardPostStopIntervalCount

Declaration location: reusable library

Lean declaration body

```
type:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P →
(Ω → NNReal) → NNReal → Ω → ℕ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
(H : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P)
(τ : Ω → NNReal) (u : NNReal) (a : Ω) =>
H.intervalCount (τ a) (τ a + u) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw.intervalCount](#)

AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (start endTime :  $\mathbb{R}$ ) (jumpTime :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (count :  $\mathbb{N}$ ) =>
  endTime - AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint start jumpTime count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.jumpTimelineEndpoint](#)

AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail

Declaration location: reusable library

Lean declaration body

```
type:
AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline  $\rightarrow \mathbb{R}$ 

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) =>
  AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes T.window.startTime T.window.endTime T.jumpTime T.count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.terminalTailFromJumpTimes](#)

AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood

Declaration location: reusable library

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ 

value:
fun (y rate exposure :  $\mathbb{R}$ ) (count :  $\mathbb{N}$ ) =>
  if count = 0 then y + (1 - y) * AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure 0
  else (1 - y) * AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R} \rightarrow \text{Prop}$ 

value:
fun (a b :  $\mathbb{R}$ ) (a_1 :  $\mathbb{R} \times \mathbb{R}$ ) => ((fun (z :  $\mathbb{R} \times \mathbb{R}$ ) => z.1 + z.2)  $^{-1}$ ) Set.loc a b) a_1
```

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount

Declaration location: paper-local

Lean declaration body

```
type:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → NNReal → Ω → ℕ

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
(M : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P) (t : NNReal) (a : Ω) =>
M.markedBirths.finiteCount (LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet 0 ↑t) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonCountingMeasureByLaw](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.firstReportSet](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ENNReal

value:
fun (rate a : ℝ) => (Set.lci 0).indicator (fun (tail : ℝ) => ENNReal.ofReal (Real.exp (- (rate * tail)))) a
```

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile

Declaration location: paper-local

Lean declaration body

```
field startDensity:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

field startDensity_nonnegative:
∀ {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T), 0 ≤ self.startDensity

field endKernel:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T →
(j : Fin (T.count + 1)) → ProbabilityTheory.Kernel (Fin ↑j → ℝ) ℝ

field endKernel_isMarkov:
∀ {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
ProbabilityTheory.IsMarkovKernel (self.endKernel j)

field endDensity:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → (j : Fin (T.count + 1)) → (Fin ↑j → ℝ) → ℝ → ENNReal

field endDensity_measurable:
∀ {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
Measurable (Function.uncurry (self.endDensity j))

field endKernel_eq_withDensity:
∀ {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (j : Fin (T.count + 1)),
self.endKernel j = (ProbabilityTheory.Kernel.const (Fin ↑j → ℝ) MeasureTheory.volume).withDensity (self.endDensity j)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)

LBG24SpatialUnderreporting.Lemma2ForwardSourceModel

Declaration location: paper-local

Lean declaration body

```

field rate:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : StandardBorelSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        [inst_2 : MeasureTheory.IsProbabilityMeasure P] → LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P → ℝ

field rate_pos:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  0 < self.rate

field interarrivalPath:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : StandardBorelSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
          LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P → Ω → ℕ → ℝ

field interarrivalPath_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  Measurable self.interarrivalPath

field interarrivalPath_hasLaw:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  ProbabilityTheory.HasLaw self.interarrivalPath
  (AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure self.rate) P

field startTime:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : StandardBorelSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
          LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P → Ω → NNReal

field observationHorizon:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : StandardBorelSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
          LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P → NNReal

field rateFreeStartKernel:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    [inst_1 : StandardBorelSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
          LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P → ProbabilityTheory.Kernel NNReal NNReal

field rateFreeStartKernel_isMarkov:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  ProbabilityTheory.IsMarkovKernel self.rateFreeStartKernel

field startTime_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  Measurable self.startTime

field firstReport_le_start:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P)
  (ω : Ω),
  (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (self.interarrivalPath ω)).toNNReal ≤
  self.startTime ω

field start_le_observationHorizon_of firstReport_le_start:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P)
  (ω : Ω),
  (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (self.interarrivalPath ω)).toNNReal ≤
  self.observationHorizon →
  self.startTime ω ≤ self.observationHorizon

field start_conditional_law:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
  ∀m (ω : Ω) ∂P,
  ProbabilityTheory.HasLaw self.startTime
  (self.rateFreeStartKernel
    (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (self.interarrivalPath ω)).toNNReal)
  ((ProbabilityTheory.condExpKernel P
    (MeasurableSpace.comap
      (fun (ω : Ω) =>
        (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (self.interarrivalPath ω)).toNNReal)
      inferInstance))
  ω)

field rateFreeStartKernel_supported_after_first:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P)
  (t : NNReal), (self.rateFreeStartKernel t) (Set.Ici t) = 1

```

```

field start_condIndep_postFirstReportTail:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
[inst_2 : MeasureTheory.IsProbabilityMeasure P] (self : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P),
ProbabilityTheory.CondIndepFun
(MeasurableSpace.comap
(fun (ω : Ω) =>
(AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (self.interarrivalPath ω)).toNNReal)
inferInstance)
(Measurable.comap_le
(Measurable.comp
(Measurable.comp measurable_real toNNReal
(AppliedModelingLib.Probability.PoissonProcess.measurable_interarrival 0))
self.interarrivalPath_measurable))
self.startTime
(fun (ω : Ω) => AppliedModelingLib.Probability.PoissonProcess.futureInterarrival 1 (self.interarrivalPath ω)) P

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure](#)
- [AppliedModelingLib.Probability.PoissonProcess.futureInterarrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.interarrival](#)

LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process

Declaration location: paper-local

Lean declaration body

```

type:
{Ω : Type u_1} →
[inst : MeasurableSpace Ω] →
[inst_1 : StandardBorelSpace Ω] →
{P : MeasureTheory.Measure Ω} →
[inst_2 : MeasureTheory.IsProbabilityMeasure P] →
LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P →
AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P

value:
fun {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
[inst_2 : MeasureTheory.IsProbabilityMeasure P] (M : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P) =>
let μ : MeasureTheory.Measure (ℕ → ℝ) :=
AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure M.rate;
let H :
AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw (ℕ → ℝ)
(AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure M.rate) :=
AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw M.rate_pos;
have hpath : MeasureTheory.MeasurePreserving M.interarrivalPath P μ :=
LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_1 M;
{ isProbability := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_2, rate := M.rate,
rate_pos := M.rate_pos, count := fun (t : NNReal) => H.count t • M.interarrivalPath,
count_measurable := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_3 M,
count_zero_ae := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_4 M,
count_mono_ae := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_5 M,
hasIndepIncrements := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_6 M,
increment_hasLaw := @LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process._proof_8 Ω inst inst_1 P inst_2 M }

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [AppliedModelingLib.Probability.PoissonProcess.canonicalForwardHomogeneousPoissonCountingProcessByLaw](#)
- [AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure](#)
- [AppliedModelingLib.Probability.PoissonProcess.rateExposureParam](#)
- [LBG24SpatialUnderreporting.Lemma2ForwardSourceModel](#)

LBG24SpatialUnderreporting.Theorem2ConditionOneSelection

Declaration location: paper-local

Lean declaration body

```

field firstReportTime:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → NNReal

field startTime:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → NNReal

field postFirstReportTail:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → Ω → Tail

field rateFreeStartKernel:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail → ProbabilityTheory.Kernel NNReal NNReal

field rateFreeStartKernel_isMarkov:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail),
  ProbabilityTheory.IsMarkovKernel self.rateFreeStartKernel

field firstReportTime_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.firstReportTime

field startTime_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.startTime

field postFirstReportTail_measurable:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail), Measurable self.postFirstReportTail

field firstReport_le_start:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail) (ω : Ω),
  self.firstReportTime ω ≤ self.startTime ω

field start_conditional_law:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail),
  ∀m (ω : Ω) ∂P,
  ProbabilityTheory.HasLaw self.startTime (self.rateFreeStartKernel (self.firstReportTime ω))
  ((ProbabilityTheory.condExpKernel P (MeasurableSpace.comap self.firstReportTime inferInstance)) ω)

field rateFreeStartKernel_supported_after_first:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail) (t : NNReal),
  (self.rateFreeStartKernel t) (Set.Ici t) = 1

field start_condIndep_postFirstReportTail:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] [inst_1 : StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2} [inst_3 : MeasurableSpace Tail]
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail),
  ProbabilityTheory.CondIndepFun (MeasurableSpace.comap self.firstReportTime inferInstance)
  (Measurable.comap_le self.firstReportTime_measurable) self.startTime self.postFirstReportTail P

```

LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel

Declaration location: paper-local

Lean declaration body

field gamma:
 LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel $\rightarrow \mathbb{R}$

field gamma_nonnegative:
 $\forall$ (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel), $0 \leq \text{self.gamma}$

field gamma_le_one:
 $\forall$ (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel), $\text{self.gamma} \leq 1$

field rate:
 LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel $\rightarrow \mathbb{R}$

field rate_pos:
 $\forall$ (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel), $0 < \text{self.rate}$

field exposure:
 LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel $\rightarrow \mathbb{R}$

field exposure_pos:
 $\forall$ (self : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel), $0 < \text{self.exposure}$

LBG24SpatialUnderreporting.assumption_lemma2_condition_one_selection

Declaration location: paper-local

Lean declaration body

```
type:
  (Ω : Type u_1) →
  [inst : MeasurableSpace Ω] →
  [StandardBorelSpace Ω] →
  (P : MeasureTheory.Measure Ω) →
  [MeasureTheory.IsProbabilityMeasure P] → (Tail : Type u_2) → [MeasurableSpace Tail] → Type (max u_1 u_2)

value:
fun (Ω : Type u_1) [MeasurableSpace Ω] [StandardBorelSpace Ω] (P : MeasureTheory.Measure Ω)
  [MeasureTheory.IsProbabilityMeasure P] (Tail : Type u_2) [MeasurableSpace Tail] =>
  LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Ω P Tail
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)

LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.selection

Declaration location: paper-local

Lean declaration body

```
type:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  [inst_1 : StandardBorelSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P →
  LBG24SpatialUnderreporting.assumption_lemma2_condition_one_selection Ω P (ℕ → ℝ)

value:
fun {Ω : Type u_1} [MeasurableSpace Ω] [StandardBorelSpace Ω] {P : MeasureTheory.Measure Ω}
  [MeasureTheory.IsProbabilityMeasure P] (M : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Ω P) =>
  {
    firstReportTime := fun (ω : Ω) =>
      (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (M.interarrivalPath ω)).toNNReal,
    startTime := M.startTime,
    postFirstReportTail := fun (ω : Ω) =>
      AppliedModelingLib.Probability.PoissonProcess.futureInterarrival 1 (M.interarrivalPath ω),
    rateFreeStartKernel := M.rateFreeStartKernel, rateFreeStartKernel_isMarkov := M.rateFreeStartKernel_isMarkov,
    firstReportTime_measurable := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.selection._proof_1 M,
    startTime_measurable := M.startTime_measurable,
    postFirstReportTail_measurable := LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.selection._proof_2 M,
    firstReport_le_start := M.firstReport_le_start, start_conditional_law := M.start_conditional_law,
    rateFreeStartKernel_supported_after_first := M.rateFreeStartKernel_supported_after_first,
    start_condIndep_postFirstReportTail := M.start_condIndep_postFirstReportTail }
  }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.futureInterarrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.interarrival](#)
- [LBG24SpatialUnderreporting.Lemma2ForwardSourceModel](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)
- [LBG24SpatialUnderreporting.assumption_lemma2_condition_one_selection](#)

LBG24SpatialUnderreporting.continuousDurationNoReportProbability

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{R} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ 

value:
fun (reportingRate :  $\mathbb{R}$ ) (durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
   $\int (t : \mathbb{R}) \text{ in } \text{Set.Ici } 0, \text{ durationDensity } t * \text{AppliedModelingLib.Probability.PoissonProcess.noArrivalProb reportingRate } t$ 
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.noArrivalProb](#)

LBG24SpatialUnderreporting.continuousDurationFirstReportProbability

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow (\mathbb{R} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}$ 

value:
fun (reportingRate :  $\mathbb{R}$ ) (durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  1 - LBG24SpatialUnderreporting.continuousDurationNoReportProbability reportingRate durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationNoReportProbability](#)

LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel.observedUniqueIncidentCount

Declaration location: paper-local

Lean declaration body

```
type:
{ $\Omega$  : Type u_1} →
[inst : MeasurableSpace  $\Omega$ ] →
{P : MeasureTheory.Measure  $\Omega$ } →
  LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel  $\Omega$  P →  $\mathbb{N} \rightarrow \Omega \rightarrow \mathbb{N}$ 

value:
fun { $\Omega$  : Type u_1} [MeasurableSpace  $\Omega$ ] {P : MeasureTheory.Measure  $\Omega$ }
  (M : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel  $\Omega$  P) (n :  $\mathbb{N}$ ) ( $\omega$  :  $\Omega$ ) =>
   $\sum i \in \text{Finset.range } n, (M.\text{calendarFirstReports}.\text{calendarCount } (\uparrow i + 1) \omega - M.\text{calendarFirstReports}.\text{calendarCount } (\uparrow i) \omega)$ 
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel](#)

LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{R} \rightarrow \mathbb{R}$ ) → ( $\mathbb{R} \rightarrow \mathbb{R}$ ) →  $\mathbb{R}$ 

value:
fun (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
   $\int (t : \mathbb{R}) \text{ in } \text{Set.Ici } 0, \text{ durationDensity } t * \text{Real.exp } (-\text{cumulativeIntensity } t)$ 
```

LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$   $\mathbb{R}$ 

value:
fun (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  1 -
    LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity cumulativeIntensity
    durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationNoReportProbabilityOfCumulativeIntensity](#)

LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity

Declaration location: paper-local

Lean declaration body

```
type:
 $\mathbb{R} \rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$  ( $\mathbb{R} \rightarrow \mathbb{R}$ )  $\rightarrow$   $\mathbb{R}$ 

value:
fun (incidentRate :  $\mathbb{R}$ ) (cumulativeIntensity durationDensity :  $\mathbb{R} \rightarrow \mathbb{R}$ ) =>
  incidentRate *
    LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity cumulativeIntensity
    durationDensity
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity](#)

LBG24SpatialUnderreporting.finiteGapPrefix

Declaration location: paper-local

Lean declaration body

```
type:
{count :  $\mathbb{N}$ }  $\rightarrow$  (Fin count  $\rightarrow$   $\mathbb{R}$ )  $\rightarrow$  (j : Fin (count + 1))  $\rightarrow$  Fin  $\uparrow$  j  $\rightarrow$   $\mathbb{R}$ 

value:
fun {count :  $\mathbb{N}$ } (gaps : Fin count  $\rightarrow$   $\mathbb{R}$ ) (j : Fin (count + 1)) (a : Fin  $\uparrow$  j) =>
  gaps (↑ a, LBG24SpatialUnderreporting.finiteGapPrefix._proof_1 j a)
```

LBG24SpatialUnderreporting.finiteArrivalPrefix

Declaration location: paper-local

Lean declaration body

```
type:
{count :  $\mathbb{N}$ }  $\rightarrow$  (Fin count  $\rightarrow$   $\mathbb{R}$ )  $\rightarrow$  (j : Fin (count + 1))  $\rightarrow$  Fin  $\uparrow$  j  $\rightarrow$   $\mathbb{R}$ 

value:
fun {count :  $\mathbb{N}$ } (gaps : Fin count  $\rightarrow$   $\mathbb{R}$ ) (j : Fin (count + 1)) (a : Fin  $\uparrow$  j) =>
  AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector (↑ j)
  (LBG24SpatialUnderreporting.finiteGapPrefix gaps j) a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.cumulativeArrivalVector](#)
- [LBG24SpatialUnderreporting.finiteGapPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel

Declaration location: paper-local

Lean declaration body

```
field startWeight:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ENNReal

field endKernel:
{count : ℕ} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count →
  (j : Fin (count + 1)) → ProbabilityTheory.Kernel (Fin ↑j → ℝ) ℝ

field endKernel_isMarkov:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  ProbabilityTheory.IsMarkovKernel (self.endKernel j)

field endDensity:
{count : ℕ} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count →
  (j : Fin (count + 1)) → (Fin ↑j → ℝ) → ℝ → ENNReal

field endDensity_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  Measurable (Function.uncurry (self.endDensity j))

field endKernel_eq_withDensity:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (j : Fin (count + 1)),
  self.endKernel j = (ProbabilityTheory.Kernel.const (Fin ↑j → ℝ) MeasureTheory.volume).withDensity (self.endDensity j)

field stageSurvival_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (i : Fin count),
  Measurable fun (gaps : Fin count → ℝ) =>
    ((self.endKernel i.castSucc) (LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc)) (Set.loi (gaps i))

field terminalDensity_measurable:
∀ {count : ℕ} (self : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count),
  Measurable fun (p : (Fin count → ℝ) × ℝ) =>
    self.endDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} →
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count) →
  M.EndpointDensityPresentation → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count)
  (D : M.EndpointDensityPresentation) =>
  { startWeight := M.selectedStartLikelihood, endKernel := M.endpointResponse,
    endKernel_isMarkov := M.endpointResponse_isMarkov, endDensity := D.endpointDensity,
    endDensity_measurable := D.endpointDensity_measurable,
    endKernel_eq_withDensity := D.endpointResponse_eq_withDensity,
    stageSurvival_measurable := M.responseSurvival_measurable,
    terminalDensity_measurable := D.terminalDensity_measurable }
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.stageSurvival

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) → Fin count → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (gaps : Fin count → ℝ)
  (i : Fin count) =>
  ((M.endKernel i.castSucc) (LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc)) (Set.loi (gaps i))
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalDensity

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (p : (Fin count → ℝ) × ℝ) =>
  M.endDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.endpointWeight

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (p : (Fin count → ℝ) × ℝ) =>
  (M.startWeight * ∏ i : Fin count, M.stageSurvival p.1 i) * M.terminalDensity p
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.stageSurvival](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalDensity](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalWeight

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ℝ → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (rate : ℝ)
  (p : (Fin count → ℝ) × ℝ) =>
  M.endpointWeight p * LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail rate p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.endpointWeight](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalNoArrivalTail](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.rawGapTailDensity

Declaration location: paper-local

Lean declaration body

```
type:
{count : ℕ} → LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count → ℝ → (Fin count → ℝ) × ℝ → ENNReal

value:
fun {count : ℕ} (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel count) (rate : ℝ)
  (p : (Fin count → ℝ) × ℝ) =>
  AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity rate count p.1 * M.terminalWeight rate p
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.exponentialBlockDensity](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.terminalWeight](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count → ℝ → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count) (rate : ℝ) =>
  (M.rawGapTailDensity rate (T.gap, T.tail)).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.rawGapTailDensity](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) →
  M.EndpointDensityPresentation → ℝ → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
  (D : M.EndpointDensityPresentation) (rate : ℝ) =>
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood T
  (LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel M
  D)
  rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.theorem2CausalConditionalLikelihood](#)

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile

Declaration location: paper-local

Lean declaration body

```
type:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T

value:
fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (M : LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel T.count) =>
  { startDensity := M.startWeight.toReal,
    startDensity_nonnegative :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_1 M,
    endKernel := M.endKernel,
    endKernel_isMarkov :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_2 M,
    endDensity := M.endDensity,
    endDensity_measurable :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_3 M,
    endKernel_eq_withDensity :=
      LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile._proof_5 M }
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) → (j : Fin (T.count + 1)) → Fin ↑j → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) (j : Fin (T.count + 1))
  (a : Fin ↑j) =>
  LBG24SpatialUnderreporting.finiteArrivalPrefix T.gap j a
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.stageSurvival

Declaration location: paper-local

Lean declaration body

```
type:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → Fin T.count → ℝ

value:
fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) (i : Fin T.count) =>
  (((M.endKernel i).castSucc) (LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix T i.castSucc)
    (Set.loi (T.gap i))).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.gap](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observedPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.terminalEndpointDensity

Declaration location: paper-local

Lean declaration body

```
type:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
  (M.endDensity (Fin.last T.count)
   (LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observationsPrefix T (Fin.last T.count)) T.tail).toReal
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline.tail](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.observationsPrefix](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.endpointWeight

Declaration location: paper-local

Lean declaration body

```
type:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
  M.startDensity * M.terminalEndpointDensity * ∏ i : Fin T.count, M.stageSurvival i
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.stageSurvival](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.terminalEndpointDensity](#)

LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.correctedResidual

Declaration location: paper-local

Lean declaration body

```
type:
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T → ℝ

value:
fun {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (M : LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile T) =>
  M.endpointWeight * (↑ T.count.factorial / T.window.exposure ^ T.count)
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.endpointWeight](#)

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual

Declaration location: paper-local

Lean declaration body

```
type:
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline) →
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) → M.EndpointDensityPresentation → ℝ

value:
fun (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
  (M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
  (D : M.EndpointDensityPresentation) =>
(LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel M
 D).toFiniteStageCausalEndpointProfile.correctedResidual
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation.toDensityModel](#)
- [LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.toFiniteStageCausalEndpointProfile](#)
- [LBG24SpatialUnderreporting.FiniteStageCausalEndpointProfile.correctedResidual](#)

LBG24SpatialUnderreporting.mleRate

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ

value:
fun (totalCount : ℕ) (totalExposure : ℝ) => ↑totalCount / totalExposure
```

LBG24SpatialUnderreporting.poissonRateLogLikelihood

Declaration location: paper-local

Lean declaration body

```
type:
ℕ → ℝ → ℝ → ℝ

value:
fun (totalCount : ℕ) (totalExposure rate : ℝ) =>
AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel totalCount totalExposure rate
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.poissonRateLogLikelihoodKernel](#)

LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain

Declaration location: paper-local

Lean declaration body

```
type:
ℝ → ℝ → ℕ → ℝ

value:
fun (rate exposure : ℝ) (count : ℕ) => AppliedModelingLib.Probability.PoissonProcess.countLikelihood rate exposure count
```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)

LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood

Declaration location: paper-local

Lean declaration body

type:

$\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$

value:

fun (gamma rate exposure : $\mathbb{R}$) (count : $\mathbb{N}$) =>

AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood gamma rate exposure count

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.zeroInflatedCountLikelihood](#)

Material library prerequisites

AppliedModelingLib.Probability.PoissonProcess.rateExposureParam

Source locator: cited publication, lines 291-323

Verbatim paper-source connection

Then, the likelihood of the data D_i as a function of the reporting rate λ , conditional on all reports before s_i , can be decomposed as

$$L(\lambda|D_i) = p$$

~

$$M_i; \lambda(e_i - s_i)$$

~

$$f(D_i),$$

(1)

where

p

~

$$M_i; \lambda(e_i - s_i)$$

~

$$= [\lambda(e_i - s_i)]$$

$$M_i$$

~

$$M_i!$$

$$\exp(-\lambda(e_i - s_i))$$

(2)

$$5$$

[form-feed in source extraction]

is the probability mass function of a Poisson distributed random variable with M_i occurrences and rate parameter $\lambda(e_i - s_i)$, and $f(D_i)$ is a function that does not depend on λ .

[Next verbatim source excerpt]

As reference, the likelihood of a Poisson distribution with $M \geq 0$ occurrences and rate $\lambda(e - s) > 0$ is as follows:

$$p(M|\lambda(e - s)) = [\lambda(e - s)]^M$$

M!

$$\exp(-\lambda(e - s)).$$

Lean-expanded library semantic target

type:
(rate exposure : $\mathbb{R}$) $\rightarrow 0 \leq \text{rate} * \text{exposure} \rightarrow \text{NNReal}$

value:
fun (rate exposure : $\mathbb{R}$) (h_nonneg : $0 \leq \text{rate} * \text{exposure}$) => (rate * exposure, h_nonneg)

Exact Lean library declaration

```
-- Nonnegative rate-exposure product as the ` $\mathbb{R}_{\geq 0}$ ` parameter of a Poisson law. -/  
def rateExposureParam (rate exposure :  $\mathbb{R}$ )  
  (h_nonneg :  $0 \leq \text{rate} * \text{exposure}$ ) :  $\mathbb{R}_{\geq 0}$  :=  
  (rate * exposure, h_nonneg)
```

Recorded source-to-library screening Verdict: matches

Reason: The source Poisson parameter is $\lambda(e-s)$. The target constructs precisely that rate-exposure product as a nonnegative real parameter, and displayed uses supply positive rate and nonnegative interval exposure.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw

Source locator: cited publication, lines 228-230

Verbatim paper-source connection

Model part 2: reports. An incident i of type θ is active in time window $[t_i, t_i + T_i]$. During this time, reports are submitted according to a homogeneous Poisson process, with rate $\lambda_\theta > 0$. Thus, the number of reports M_i for incident i follows a Poisson distribution, $M_i \sim \text{Poisson}(\lambda_\theta T_i)$.

[Next verbatim source excerpt]

Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random variables of the start and end of an observation period S and E , and suppose S and E satisfy the following conditions, respectively:

Lean declaration type (metadata)

```
field isProbability:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
    MeasureTheory.IsProbabilityMeasure P

field rate:
  {Ω : Type u_1} →
    [inst : MeasurableSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P → ℝ

field rate_pos:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
    0 < self.rate

field count:
  {Ω : Type u_1} →
    [inst : MeasurableSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P → NNReal → Ω → ℕ

field count_measurable:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P) (t : NNReal),
    Measurable (self.count t)

field count_zero_ae:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
    ∀m (ω : Ω) ∂P, self.count 0 ω = 0

field count_mono_ae:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
    ∀m (ω : Ω) ∂P, Monotone fun (t : NNReal) => self.count t ω

field hasIndepIncrements:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P),
    ProbabilityTheory.HasIndepIncrements self.count P

field increment_hasLaw:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Ω P)
    {s t : NNReal} (hst : s ≤ t),
    ProbabilityTheory.HasLaw (fun (ω : Ω) => self.count t ω - self.count s ω)
      (ProbabilityTheory.poissonMeasure
        (AppliedModelingLib.Probability.PoissonProcess.rateExposureParam self.rate (↑t - ↑s)
          (mul_nonneg (le_of_lt self.rate_pos) (sub_nonneg.mpr (NNReal.coe_le_coe.mpr hst))))))
  P
```

Exact Lean library declaration

/--
A positive-rate homogeneous Poisson counting process on a forward time axis.

The probability-law field deliberately makes this suitable for a stationary/Palm law once the caller separately constructs that law. It says nothing about how that Palm law was obtained. `HasIndepIncrements` is the mathlib process predicate, specialized here to $\mathbb{R}_{\geq 0}$ time.

-/

```
structure ForwardHomogeneousPoissonCountingProcessByLaw
  (Ω : Type*) [MeasurableSpace Ω] (P : Measure Ω) where
  /-- The forward process is carried by a probability law. -/
  isProbability : IsProbabilityMeasure P
  rate : ℝ
  rate_pos : 0 < rate
  count : ℝ≥0 → Ω → ℕ
  count_measurable : ∀ t, Measurable (count t)
  count_zero_ae : ∀m ω ∂P, count 0 ω = 0
  count_mono_ae : ∀m ω ∂P, Monotone fun t => count t ω
  hasIndepIncrements : ProbabilityTheory.HasIndepIncrements count P
  increment_hasLaw :
```

```

∀ {s t : ℝ≥0} (hst : s ≤ t),
  ProbabilityTheory.HasLaw
    (fun ω : Ω => count t ω - count s ω)
    (ProbabilityTheory.poissonMeasure
      (rateExposureParam rate ((t : ℝ) - (s : ℝ))
        (mul_nonneg (le_of_lt rate_pos)
          (sub_nonneg.mpr (NNReal.coe_le_coe.mpr hst))))) P

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.rateExposureParam](#)

Recorded source-to-library screening Verdict: matches

Reason: The source requires a positive-rate homogeneous Poisson counting process. The target has positive rate, zero initial count, monotone paths, independent increments, and Poisson increment law with parameter $\text{rate} \cdot (t-s)$; its displayed constructions and uses remain on nonnegative forward time.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-specific semantic prerequisites

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel

Source locator: cited publication, lines 1848-1856

Verbatim paper-source connection

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some $m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is

$P[E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}] = h_m(t)$, for some $m \in \mathbb{Z}^+$, $\forall T \geq t \geq t_m$,

where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S :

$P[E = t | T_1 = t_1, S = s] = P[E = t | T_1 = t_1]$, $\forall T \geq t \geq t_1, \forall s \geq t_1$.

Lean-elaborated paper signature

field selectedStartLikelihood:

$\{\text{count} : \mathbb{N}\} \rightarrow \text{LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count} \rightarrow \text{ENNReal}$

field endpointResponse:

$\{\text{count} : \mathbb{N}\} \rightarrow$
 $\text{LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count} \rightarrow$
 $(j : \text{Fin (count + 1)}) \rightarrow \text{ProbabilityTheory.Kernel (Fin } \uparrow j \rightarrow \mathbb{R}) \mathbb{R}$

field endpointResponse_isMarkov:

$\forall \{\text{count} : \mathbb{N}\} (\text{self} : \text{LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}) (j : \text{Fin (count + 1)})$
 $\text{ProbabilityTheory.IsMarkovKernel (self.endpointResponse j)}$

field endpointResponse_nonnegative_support:

$\forall \{\text{count} : \mathbb{N}\} (\text{self} : \text{LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}) (j : \text{Fin (count + 1)})$
 $(\text{history} : \text{Fin } \uparrow j \rightarrow \mathbb{R}), ((\text{self.endpointResponse j}) \text{ history}) (\text{Set.lim 0}) = 0$

field responseSurvival_measurable:

$\forall \{\text{count} : \mathbb{N}\} (\text{self} : \text{LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}) (i : \text{Fin count})$
 $\text{Measurable fun (gaps : Fin count} \rightarrow \mathbb{R}) =>$
 $((\text{self.endpointResponse i.castSucc}) (\text{LBG24SpatialUnderreporting.finiteArrivalPrefix gaps i.castSucc}))$
 $(\text{Set.lim (gaps i)})$

Direct retained dependencies

- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Condition 2 makes the endpoint law depend only on the report prefix visible at the current time and not on the reporting rate or selected start. The target uses a rate-free Markov kernel on exactly that visible finite arrival prefix, with nonnegative response support and the displayed fixed-model likelihood use.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation

Source locator: cited publication, lines 1848-1856

Verbatim paper-source connection

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some $m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is

$P[E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}] = h_m(t)$, for some $m \in \mathbb{Z}^+$, $\forall T \geq t \geq t_m$,

where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S :

$P[E = t | T_1 = t_1, S = s] = P[E = t | T_1 = t_1]$, $\forall T \geq t \geq t_1, \forall s \geq t_1$.

Lean-elaborated paper signature

field endpointDensity:

```
{count : ℕ} →
  {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count} →
    M.EndpointDensityPresentation → (j : Fin (count + 1)) → (Fin ↑ j → ℝ) → ℝ → ENNReal
```

field endpointDensity_measurable:

```
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
  (self : M.EndpointDensityPresentation) (j : Fin (count + 1)), Measurable (Function.uncurry (self.endpointDensity j))
```

field endpointResponse_eq_withDensity:

```
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
  (self : M.EndpointDensityPresentation) (j : Fin (count + 1)),
    M.endpointResponse j =
      (ProbabilityTheory.Kernel.const (Fin ↑ j → ℝ) MeasureTheory.volume).withDensity (self.endpointDensity j)
```

field terminalDensity_measurable:

```
∀ {count : ℕ} {M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel count}
  (self : M.EndpointDensityPresentation),
    Measurable fun (p : (Fin count → ℝ) × ℝ) =>
      self.endpointDensity (Fin.last count) (LBG24SpatialUnderreporting.finiteArrivalPrefix p.1 (Fin.last count)) p.2
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.finiteArrivalPrefix](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The source writes the conditional endpoint law as $h_m(t)$. The target supplies a measurable h_m -style density for each visible prefix and identifies the endpoint kernel with Lebesgue measure weighted by that density, matching the density branch used by the displayed Equation (8) likelihood.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.CalendarFirstReportSourceModel

Source locator: cited publication, lines 1660-1770

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\text{observed}}^\lambda$ follows a Poisson process with rate λ' , where

$$\lambda' = \lambda \theta$$

$$\theta = \frac{1}{\int_0^\infty \exp(-\lambda \theta t) \lambda \theta f(t) dt}$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\lambda' = \lambda \theta = \lambda \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\lambda \theta t)$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty \Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^\infty (1 - \exp(-\lambda \theta t)) f(t) dt = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

$$\int_0^t \lambda \theta(u) du$$

$$\int_0^t f(t) dt,$$

which simplifies under time-homogeneity to

$$p = 1 - \exp(-\lambda \theta t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_\theta(t)$ follows a Poisson process with rate $\lambda \theta$. Conditional on $N_\theta(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\text{observed}}^\theta(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\text{observed}}^\theta(t)$ follows a Poisson distribution with rate $\lambda \theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

```

field incidentRate:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → ℝ

field incidentRate_nonnegative:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), 0 ≤ self.incidentRate

field retentionProbability:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → ℝ

field retentionProbability_nonnegative:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), 0 ≤ self.retentionProbability

field retentionProbability_le_one:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), self.retentionProbability ≤ 1

field retainedDelay:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P → MeasureTheory.Measure ℝ

field retainedDelay_finite:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P),
  MeasureTheory.IsFiniteMeasure self.retainedDelay

field retainedDelay_nonnegative_support:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P), self.retainedDelay (Set.liv 0) = 0

field retainedDelay_mass:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P),
  self.retainedDelay Set.univ = ENNReal.ofReal self.retentionProbability

field markedBirths:
  {Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  (self : LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P) →
  AppliedModelingLib.Probability.PoissonCountingMeasureByLaw Ω (ℝ × ℝ) P
  (ENNReal.ofReal self.incidentRate • MeasureTheory.volume.prod self.retainedDelay)

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonCountingMeasureByLaw](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Under the approved calendar-first-report reading, observed incidents are calendar-time first reports. The target's stationary marked Poisson births, nonnegative retained delays, and retained-delay mass p yield first-report intensity $\text{Lambda} * p$ and support the displayed Lemma 1 and Proposition 1 uses.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

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Paper-local declaration:

LBG24SpatialUnderreporting.CollapsedFiniteStageEndpointModel.iidGapLaw

Source locator: cited publication, lines 228-230

Verbatim paper-source connection

Model part 2: reports. An incident i of type θ is active in time window $[t_i, t_i + T_i]$. During this time, reports are submitted according to a homogeneous Poisson process, with rate $\lambda_\theta > 0$. Thus, the number of reports M_i for incident i follows a Poisson distribution, $M_i \sim \text{Poisson}(\lambda_\theta T_i)$.

[Next verbatim source excerpt]

Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random variables of the start and end of an observation period S and E , and suppose S and E satisfy the following conditions, respectively:

Lean-expanded paper semantic target

type:

$(\text{count} : \mathbb{N}) \rightarrow \mathbb{R} \rightarrow \text{MeasureTheory.Measure } (\text{Fin count} \rightarrow \mathbb{R})$

value:

$\text{fun } (\text{count} : \mathbb{N}) (\text{rate} : \mathbb{R}) \Rightarrow \text{MeasureTheory.Measure.pi fun } (x : \text{Fin count}) \Rightarrow \text{ProbabilityTheory.expMeasure rate}$

Recorded source-to-declaration screening Verdict: matches

Reason: The source assumes a positive-rate homogeneous Poisson report process; on that domain its finite interarrival blocks have the target's product law of iid exponential gaps. The target's additional out-of-domain rate inputs do not alter positive-rate behavior.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition

Paper-local declaration:

LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition

Source locator: cited publication, lines 1660-1770

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\text{observed}}^\Lambda$ follows a Poisson process with rate Λ' , where

$$\Lambda' = \Lambda \theta$$

$$= \int_0^\infty \frac{1 - \exp(-\lambda \theta u)}{\theta} f(u) du.$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda' = \Lambda \theta \int_0^\infty \exp(-\lambda \theta t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\lambda \theta t)$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty \Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^\infty (1 - \exp(-\lambda \theta t)) f(t) dt = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

—
t

0
 $\lambda\theta(u)du$
f(t)dt,

which simplifies under time-homogeneity to

$p = 1 -$
 $\int_0^\infty$

0
 $\exp(-\lambda\theta t) f(t)dt.$

Over a time interval of t, the total number of incidents of type θ that happen, $N\theta(t)$ follows a Poisson process with rate $\Lambda\theta$.
Conditional on $N\theta(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\text{observed}}\theta(t)$ follows a binomial distribution with parameters (n, p) .
Thus $N_{\text{observed}}\theta(t)$ follows a Poisson distribution with rate $\Lambda\theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field integrable:
 $\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
MeasureTheory.Integrable durationDensity (MeasureTheory.volume.restrict (Set.Ici 0))

field nonnegative_ae:
 $\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
 $\forall^m (t : \mathbb{R}) \partial \text{MeasureTheory.volume.restrict (Set.Ici 0)}, 0 \leq \text{durationDensity } t$

field integral_eq_one:
 $\forall \{ \text{durationDensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition durationDensity $\rightarrow$
 $\int (t : \mathbb{R}) \text{ in Set.Ici 0, durationDensity } t = 1$

Recorded source-to-declaration screening Verdict: matches
Reason: The source specifies f as a probability density on [0,infinity). The target states the corresponding restricted-domain integrability, almost-everywhere nonnegativity, and unit integral, with values outside that interval semantically irrelevant.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel

Source locator: cited publication, lines 219-226

Verbatim paper-source connection

Model part 1: incident birth and death. Incidents of type θ occur (“are born”) according to a random process parameterized by $\Lambda\theta$. (For example, a homogeneous Poisson process with rate $\Lambda\theta$). Type θ encodes geographic and incident-level (e.g., whether it is a hazard) characteristics. For each incident i , let t_i denote its birth time. Furthermore, the incident “dies” at some time $t_i + T_i$, and no more reports are logged. The death time reflects, for example, the incident being resolved by the agency or by an outside group, or reports no longer being logged because the issue has been inspected. Crucially, birth times are unobserved in the model; death times may be unobserved or partially observed.⁴

Lean-elaborated paper signature

field isProbability:

```
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
  {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P),
  MeasureTheory.IsProbabilityMeasure P
```

field incidentType:

```
{IncidentType : Type u_1} →
  {Ω : Type u_2} →
  {Incident : Type u_3} →
  {Report : Type u_4} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
  Incident → IncidentType
```

field occurrenceParameter:

```
{IncidentType : Type u_1} →
  {Ω : Type u_2} →
  {Incident : Type u_3} →
  {Report : Type u_4} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
  IncidentType → ℝ → ℝ
```

field birthProcess:

```
{IncidentType : Type u_1} →
  {Ω : Type u_2} →
  {Incident : Type u_3} →
  {Report : Type u_4} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
  (IncidentType → ℝ → ℝ) → IncidentType → Ω → Incident → ℝ
```

field birthProcess_measurable:

```
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
  {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P)
  (occurrenceParameter : IncidentType → ℝ → ℝ) (incidentType : IncidentType) (incident : Incident),
  Measurable fun (ω : Ω) => self.birthProcess occurrenceParameter incidentType ω incident
```

field birthTime:

```
{IncidentType : Type u_1} →
  {Ω : Type u_2} →
  {Incident : Type u_3} →
  {Report : Type u_4} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P → Ω → Incident → ℝ
```

field birthTime_from_occurrence_process:

```
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
  {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P) (ω : Ω)
  (incident : Incident),
  self.birthTime ω incident = self.birthProcess self.occurrenceParameter (self.incidentType incident) ω incident
```

field duration:

```
{IncidentType : Type u_1} →
  {Ω : Type u_2} →
  {Incident : Type u_3} →
  {Report : Type u_4} →
  [inst : MeasurableSpace Ω] →
  {P : MeasureTheory.Measure Ω} →
  LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P → Ω → Incident → ℝ
```

field duration_nonnegative:

```
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
  {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P) (ω : Ω)
  (incident : Incident), 0 ≤ self.duration ω incident
```

field birthTimeObservable:

```
{IncidentType : Type u_1} →
```

```

{Ω : Type u_2} →
{Incident : Type u_3} →
{Report : Type u_4} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
Ω → Incident → Prop

field birthTime_unobserved:
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
{P : MeasureTheory.Measure Ω}
(self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P) (ω : Ω)
(incident : Incident), ¬self.birthTimeObservable ω incident

field deathTimeInformation:
{IncidentType : Type u_1} →
{Ω : Type u_2} →
{Incident : Type u_3} →
{Report : Type u_4} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
Ω → Incident → Option (Set ℝ)

field deathTimeInformation_sound:
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
{P : MeasureTheory.Measure Ω}
(self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P) (ω : Ω)
(incident : Incident) (candidates : Set ℝ),
self.deathTimeInformation ω incident = some candidates →
self.birthTime ω incident + self.duration ω incident ∈ candidates

field reportIncident:
{IncidentType : Type u_1} →
{Ω : Type u_2} →
{Incident : Type u_3} →
{Report : Type u_4} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P →
Report → Incident

field reportTime:
{IncidentType : Type u_1} →
{Ω : Type u_2} →
{Incident : Type u_3} →
{Report : Type u_4} →
[inst : MeasurableSpace Ω] →
{P : MeasureTheory.Measure Ω} →
LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P → Ω → Report → ℝ

field report_during_active_window:
∀ {IncidentType : Type u_1} {Ω : Type u_2} {Incident : Type u_3} {Report : Type u_4} [inst : MeasurableSpace Ω]
{P : MeasureTheory.Measure Ω}
(self : LBG24SpatialUnderreporting.IncidentBirthDeathSourceModel IncidentType Ω Incident Report P) (ω : Ω)
(report : Report),
self.birthTime ω (self.reportIncident report) ≤ self.reportTime ω report ∧
self.reportTime ω report ≤
self.birthTime ω (self.reportIncident report) + self.duration ω (self.reportIncident report)

```

Recorded source-to-declaration screening Verdict: matches

Reason: The target models a measurable type-indexed random birth process driven by a generic Lambda_theta-like parameter without imposing a Poisson law. It assigns latent birth times and nonnegative lifetimes, represents absent, exact, or partial sound death-time information through Option (Set Real), and confines logged reports to each incident's active birth-to-death window.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition

Paper-local declaration:

LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition

Source locator: cited publication, lines 1660-1770

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{observed}$ follows a Poisson process with rate Λ' , where

$$\Lambda' = \Lambda \theta$$

$$\int_0^\infty \frac{1 - \exp(-\lambda \theta t)}{\lambda \theta} f(t) dt.$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda' = \Lambda \theta \int_0^\infty \exp(-\lambda \theta t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\lambda \theta t)$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty \Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^\infty (1 - \exp(-\lambda \theta t)) f(t) dt = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

$$\begin{aligned}
 &= \int_0^t \lambda \theta(u) du \\
 &= \int_0^t f(t) dt,
 \end{aligned}$$

which simplifies under time-homogeneity to

$$p = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_\theta(t)$ follows a Poisson process with rate $\Lambda \theta$. Conditional on $N_\theta(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\text{observed}}^\theta(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\text{observed}}^\theta(t)$ follows a Poisson distribution with rate $\Lambda \theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

field nonnegative:
 $\forall \{ \text{reportingIntensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
 $\text{LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition reportingIntensity} \rightarrow$
 $\forall (t : \mathbb{R}), 0 \leq t \rightarrow 0 \leq \text{reportingIntensity } t$

field intervalIntegrable:
 $\forall \{ \text{reportingIntensity} : \mathbb{R} \rightarrow \mathbb{R} \},$
 $\text{LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition reportingIntensity} \rightarrow$
 $\forall (t : \mathbb{R}), 0 \leq t \rightarrow \text{IntervalIntegrable reportingIntensity MeasureTheory.volume } 0 \ t$

Recorded source-to-declaration screening Verdict: matches
Reason: The source’s age-varying reporting rate $\lambda_{\theta}(u)$ appears through finite cumulative intensities in Poisson parameters. Nonnegativity and finite-interval integrability are the domain conditions needed for that source intensity, and the displayed use retains the same cumulative-intensity formula.
Reviewer check: ☐ Matches source input
 If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel

Paper-local declaration:

LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel

Source locator: cited publication, lines 1660-1770

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\text{observed}}^\Lambda$ follows a Poisson process with rate Λ' , where

$$\Lambda' = \Lambda \theta$$

$$\theta = \frac{1}{\int_0^\infty \exp(-\lambda \theta u) \lambda \theta f(u) du}$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda' = \Lambda \theta = \Lambda \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\lambda \theta t)$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty \Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^\infty (1 - \exp(-\lambda \theta t)) f(t) dt = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt$$

$$-\int_0^t \lambda \theta(u) du + \int_0^t f(u) du,$$
 which simplifies under time-homogeneity to

$$p = 1 - \int_0^\infty \exp(-\lambda \theta t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_\theta(t)$ follows a Poisson process with rate $\Lambda \theta$. Conditional on $N_\theta(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\text{observed}}^\theta(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\text{observed}}^\theta(t)$ follows a Poisson distribution with rate $\Lambda \theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

```

field reportingIntensity:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P → ℝ → ℝ

field reportingIntensity_source:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
    LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition self.reportingIntensity

field durationDensity:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P → ℝ → ℝ

field durationDensity_source:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
    LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition self.durationDensity

field calendarFirstReports:
{Ω : Type u_1} →
  [inst : MeasurableSpace Ω] →
    {P : MeasureTheory.Measure Ω} →
      LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P →
        LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P

field preFirstReport_survival_retention_probability:
∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
  (self : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Ω P),
    self.calendarFirstReports.retentionProbability =
      LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity
    (fun (t : ℝ) => ∫ (u : ℝ) in 0..t, self.reportingIntensity u) self.durationDensity
  
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition](#)
- [LBG24SpatialUnderreporting.Lemma1ReportingIntensitySourceCondition](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbabilityOfCumulativeIntensity](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The target combines the source intensity and duration-density primitives with the approved stationary calendar-first-report model, and equates retained mass to one minus the integral of $\exp(-\int_0^t \lambda(u) du) f(t)$. This is the source rate formula under the approved pre-first-report survival bridge.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel

Source locator: cited publication, lines 1660-1770

Verbatim paper-source connection

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{observed}$ follows a Poisson process with rate Λ' , where

$$\Lambda' = \Lambda \theta$$

$$= \int_0^\infty \exp\left(-\int_0^t \lambda \theta(u) du\right) f(t) dt.$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda' = \Lambda \theta \int_0^\infty \exp(-\lambda \theta t) f(t) dt.$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\int_0^t \lambda \theta(u) du).$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty \Pr[m(t) \geq 1 | T_i = t] f(t) dt = \int_0^\infty (1 - \exp(-\int_0^t \lambda \theta(u) du)) f(t) dt = 1 - \int_0^\infty \exp(-\int_0^t \lambda \theta(u) du) f(t) dt.$$

$$\int_0^t \lambda \theta(u) du$$

$$f(t) dt,$$

which simplifies under time-homogeneity to

$$p = 1 - \exp(-\lambda \theta t) f(t) dt.$$

Over a time interval of t , the total number of incidents of type θ that happen, $N_\theta(t)$ follows a Poisson process with rate $\lambda \theta$. Conditional on $N_\theta(t) = n$, $n = 0, 1, \dots$, under steady state, $N_{\text{observed}}^\theta(t)$ follows a binomial distribution with parameters (n, p) . Thus $N_{\text{observed}}^\theta(t)$ follows a Poisson distribution with rate $\lambda \theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Lean-elaborated paper signature

```

field reportingRate:
  {Ω : Type u_1} →
    [inst : MeasurableSpace Ω] →
      {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P → ℝ

field reportingRate_nonnegative:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P), 0 ≤ self.reportingRate

field durationDensity:
  {Ω : Type u_1} →
    [inst : MeasurableSpace Ω] →
      {P : MeasureTheory.Measure Ω} → LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P → ℝ → ℝ

field durationDensity_source:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P),
      LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition self.durationDensity

field calendarFirstReports:
  {Ω : Type u_1} →
    [inst : MeasurableSpace Ω] →
      {P : MeasureTheory.Measure Ω} →
        LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P →
          LBG24SpatialUnderreporting.CalendarFirstReportSourceModel Ω P

field preFirstReport_survival_retention_probability:
  ∀ {Ω : Type u_1} [inst : MeasurableSpace Ω] {P : MeasureTheory.Measure Ω}
    (self : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Ω P),
      self.calendarFirstReports.retentionProbability =
        LBG24SpatialUnderreporting.continuousDurationFirstReportProbability self.reportingRate self.durationDensity

```

Direct retained dependencies

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.ContinuousDurationDensitySourceCondition](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbability](#)

Recorded source-to-declaration screening Verdict: matches

Reason: The target is the homogeneous specialization of the approved calendar-time model: reporting rate λ , duration density f , and retention probability one minus the integral of $\exp(-\lambda \theta t) f(t)$. Its displayed use varies λ inversely with that probability and obtains the same observed rate while retaining distinct reporting rates.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation

Paper-local declaration:

LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation

Source locator: cited publication, lines 1843-1847

Verbatim paper-source connection

Condition 1: Given $T_1 = t_1$, S is defined on $[t_1, T]$, and

$P(S = t | T_1 = t_1) = g(t)$, $\forall t \geq t_1$,

where $g(t)$ is not dependent on λ or the sample path of the Poisson process after t_1 .

Lean-elaborated paper signature

```
field startDensity:
  {Omega : Type u_1} →
  [inst : MeasurableSpace Omega] →
  [inst_1 : StandardBorelSpace Omega] →
  {P : MeasureTheory.Measure Omega} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail} →
  {startReference : MeasureTheory.Measure NNReal} →
  [inst_4 : MeasureTheory.SFinite startReference] →
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection
  startReference T →
  NNReal → NNReal → ENNReal

field startDensity_measurable:
  ∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
  [inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
  {startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
  Measurable (Function.uncurry self.startDensity)

field startKernel_eq_withDensity:
  ∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
  [inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
  {startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
  selection.rateFreeStartKernel = (ProbabilityTheory.Kernel.const NNReal startReference).withDensity self.startDensity

field observedState:
  {Omega : Type u_1} →
  [inst : MeasurableSpace Omega] →
  [inst_1 : StandardBorelSpace Omega] →
  {P : MeasureTheory.Measure Omega} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail} →
  {startReference : MeasureTheory.Measure NNReal} →
  [inst_4 : MeasureTheory.SFinite startReference] →
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection
  startReference T →
  Omega

field observationHorizon:
  {Omega : Type u_1} →
  [inst : MeasurableSpace Omega] →
  [inst_1 : StandardBorelSpace Omega] →
  {P : MeasureTheory.Measure Omega} →
  [inst_2 : MeasureTheory.IsProbabilityMeasure P] →
  {Tail : Type u_2} →
  [inst_3 : MeasurableSpace Tail] →
  {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail} →
  {startReference : MeasureTheory.Measure NNReal} →
  [inst_4 : MeasureTheory.SFinite startReference] →
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
  LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection
  startReference T →
  NNReal

field startKernel_supported_before_horizon:
  ∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
  [inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
  {startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
  {T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
  (self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
  ∀ t ≤ self.observationHorizon, (selection.rateFreeStartKernel t) (Set.lic self.observationHorizon) = 1

field observed_start_le_horizon:
  ∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
  [inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
```

```

{startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
selection.startTime self.observedState ≤ self.observationHorizon

field observed_start_eq_timeline_start:
∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
{P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
[inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
{startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
↑(selection.startTime self.observedState) = T.window.startTime

field selectedStartLikelihood:
{Omega : Type u_1} →
[inst : MeasurableSpace Omega] →
[inst_1 : StandardBorelSpace Omega] →
{P : MeasureTheory.Measure Omega} →
[inst_2 : MeasureTheory.IsProbabilityMeasure P] →
{Tail : Type u_2} →
[inst_3 : MeasurableSpace Tail] →
{selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail} →
{startReference : MeasureTheory.Measure NNReal} →
[inst_4 : MeasureTheory.SFinite startReference] →
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline} →
LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection
startReference T →
ENNReal

field selectedStartLikelihood_eq_density:
∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
{P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
[inst_3 : MeasurableSpace Tail] {selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail}
{startReference : MeasureTheory.Measure NNReal} [inst_4 : MeasureTheory.SFinite startReference]
{T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline}
(self : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
self.selectedStartLikelihood =
self.startDensity (selection.firstReportTime self.observedState) (selection.startTime self.observedState)

```

Direct retained dependencies

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Its dependent selection supplies an a.e. conditional start law via a rate-free kernel, lower support $S \geq T_1$, and conditional independence from the post- T_1 tail; this presentation identifies that kernel with a measurable conditional density. Its upper support is correctly guarded by $t \leq \text{observationHorizon}$, which gives $[t, \text{observationHorizon}]$ on the source domain $T \geq t \geq t_1$ without imposing an inconsistent requirement for $t > T$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.equation33_nyc_observation_end

Paper-local declaration:

LBG24SpatialUnderreporting.equation33_nyc_observation_end

Source locator: cited publication, lines 2437-2467

Verbatim paper-source connection

Constructing an observation interval $(S_i, E_i]$
As outlined in Theorem 1, we must be careful
in how we choose an observation interval $(S_i, E_i]$ in which we count reports – we need that the
interval is inside the incident lifetime, i.e., we must end the interval before the incident is addressed,
 $E_i \leq t_i + T_i$. Both endpoints of the interval must also satisfy the conditions outlined in Theorem 1.
As discussed above, the best choice to start the observation period S_i is the time of the first report,
but choosing the observation end is a design choice.
We make the following choice. Let t_{INSP}

i
be the inspection time of incident i , and, t_{WO}
 i
be the
time that a work order is placed for incident i , if applicable. Then, E_i of each incident i is:

$E_i = \min$

$\diamond$
100 days + S_i , t_{INSP}

i
, t_{WO}
 i

$\diamond$
.

(33)

The maximum duration of 100 is a design choice, for which we perform robustness checks (with 30 and 200 days); a maximum mitigates – for incidents not inspected for a long time – the inclusion of a time period in which an issue might have been resolved before an inspection, which would bias our estimates downward. Equation (33) requires that inspection and work order times are stopping times; that they do not depend on the future trivially holds, and it is likely that they do not depend on the reporting rate except through the type θ and the sample path number of reports received

Lean-expanded paper semantic target

type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$

value:
fun (startTime inspectionTime workOrderTime : $\mathbb{R}$) => min (min (startTime + 100) inspectionTime) workOrderTime

Recorded source-to-declaration screening Verdict: matches

Reason: The target is $\min(S_i + 100, \text{inspectionTime}_i, \text{workOrderTime}_i)$, matching Equation (33)'s three endpoint candidates.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.equation34_chicago_observation_end

Source locator: cited publication, lines 5610-5628

Verbatim paper-source connection

Constructing an observation interval (S_i, E_i)
For Chicago reports, we make the following
choice. Let $\tilde{t}_i$ be the first time an incident is reported, t_{CLOSED}
 i
be the time an incident is marked
closed, and $\tilde{t}$ be the time that we retrieved the dataset (and that it was last updated), 7/4/2022 at
21:40 EDT. Then, E_i of each incident i is:

$$E_i = \min \left(\begin{matrix} 100 \text{ days} + \tilde{t}_i, t_{CLOSED} \\ \tilde{t} \end{matrix} \right) \quad (34)$$

The choice of adding $\tilde{t}$ ensures that for “open” incidents, the observation interval is not overly long, which could bias our estimates for the reporting rate downward.

Lean-expanded paper semantic target

type:
 $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$
value:
 $\text{fun } (\text{firstReportTime } \text{closedTime } \text{retrievalTime} : \mathbb{R}) \Rightarrow \min (\min (\text{firstReportTime} + 100) \text{closedTime}) \text{retrievalTime}$

Recorded source-to-declaration screening Verdict: matches
Reason: The target is $\min(\text{firstReportTime}_i+100, \text{closedTime}_i, \text{retrievalTime})$, matching Equation (34)’s three endpoint candidates.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Paper-local declaration:

LBG24SpatialUnderreporting.poissonRegressionRate

Source locator: cited publication, lines 351-365

Verbatim paper-source connection

Second, we must empirically estimate reporting rates given high dimensional incident types θ . We leverage Poisson regression, i.e., we assume the form

$$\lambda\theta = \exp(\alpha + \beta^T \theta)$$

(4)

To better match the data distribution, we further fit a zero-inflation component (corresponding to the probability that we receive no reports in the given period).

To accommodate the high dimensionality of the incident type vector θ , we fit our models in a Bayesian setting and report posterior distributions of regression coefficients α and β .

[Next verbatim source excerpt]

First, we assume that the reporting rate is affected by the type vector θ as in a Poisson regression: for each model, we have a standard reporting rate $\lambda\theta$, defined as:

$$\lambda\theta = \exp(\alpha + \beta^T \theta)$$

(5)

Lean-expanded paper semantic target

```
type:
{Feature : Type u_1} → [Fintype Feature] → ℝ → (Feature → ℝ) → (Feature → ℝ) → ℝ

value:
fun {Feature : Type u_1} [Fintype Feature] (alpha : ℝ) (beta theta : Feature → ℝ) =>
  Real.exp (alpha + ∑ j : Feature, beta j * theta j)
```

Recorded source-to-declaration screening Verdict: matches

Reason: The target computes $\exp(\alpha + \sum_j \beta_j \cdot \theta_j)$, the finite-coordinate expansion of the source formulas $\lambda\theta = \exp(\alpha + \beta^T \theta)$ in Equations (4) and (5).

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood

Paper-local declaration:

LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood

Source locator: cited publication, lines 945-971

Verbatim paper-source connection

We further fit a zero-inflated Poisson regression [19], which corresponds to a data generating process in which, independently for each incident, only one report can be generated with probability γ , and thus there are 0 duplicate reports:

$M_i = 0.9$

For example, perhaps those incidents are personal requests for service (e.g., to plant a tree somewhere or to prune a tree to enable certain construction), for which duplicates are unlikely; alternatively, officials may physically mark the location of an incident as “report received; full inspection pending,” preventing future reports. Under this case, the likelihood of the data satisfies:

$L(\lambda\theta_i|D_i) \propto$

$(1 - \gamma)^p$

$\sim$

$M_i; \lambda\theta_i(e_i - s_i)$

$\sim$

$\sim$

$M_i \geq 1;$

$\gamma + (1 - \gamma)p(0; \lambda\theta_i(e_i - s_i)) ,$

$M_i = 0.$

(7)

Lean-expanded paper semantic target

```
type:
{Incident : Type u_1} →
  Finset Incident → (Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) → (Incident → ℕ) → ℝ
```

value:

```
fun {Incident : Type u_1} (incidents : Finset Incident)
  (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ) =>
  ∏ i ∈ incidents,
    LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
    (count i)
```

Direct retained dependencies

- [LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel](#)
- [LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood](#)

Recorded source-to-declaration screening Verdict: matches

Reason: Equation (7) gives $\gamma + (1 - \gamma)p(0)$ for zero duplicates and $(1 - \gamma)p(M)$ for positive duplicates, independently by incident. The target multiplies exactly those per-incident mixture likelihoods over the finite incident family.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 206-214

Verbatim source input

Figure 1: Model of resident reports of an incident i of type θ . Observed and unobserved events are marked with solid and dotted lines, respectively. In this example, there are a total of 3 reports made about this incident. When the reports are generated according to a homogeneous Poisson process with rate λ_θ , the average reporting delay is $\frac{1}{\lambda_\theta}$. Incident ‘death’ refers to the incident being resolved or otherwise marked such that no future reports are submitted and logged. The goal is to estimate the reporting rate λ_θ (and consequently, the unobserved reporting delay) for each incident of type θ .

[Next verbatim source excerpt]

Research question.
We are interested in auditing the efficiency and equity of the reporting process: how quickly are incidents reported, and how does that reporting rate λ_θ vary with the incident type θ ?
Estimating this value directly reveals heterogeneous reporting delays, as the expected time between incident occurrence and first report in the model is $\frac{1}{\lambda_\theta}$.

Expanded PaperInterface specification

$\forall (rate : \mathbb{R}) (rate_pos : 0 < rate), \int (x : \mathbb{R}), x \partial \{ rate := rate, rate_pos := rate_pos \}.measure = 1 / rate$

Retained governing declarations

- [AppliedModelingLib.Probability.Exponential.Model](#)
- [AppliedModelingLib.Probability.Exponential.Model.measure](#)

Lean-elaborated claim roles

- 1. parameter: $\mathbb{R}$
- 2. assumption: $0 < rate$
- 3. conclusion: $\int (x : \mathbb{R}), x \partial \{ rate := rate, rate_pos := rate_pos \}.measure = 1 / rate$

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceHomogeneousReportingDelayMean

Recorded source-to-Spec screening Verdict: matches
Reason: The source identifies the homogeneous-Poisson first-report delay expectation as reciprocal rate; the Spec is precisely the expectation of the rate-exponential law.
Reviewer check: ☐ Matches source input
If left unchecked, explain the discrepancy or uncertainty below.
Reviewer annotation

Review row 2

Source locator: cited publication, lines 1833-1868

Verbatim source input

Formally, we separate terms involving λ from terms independent of λ , which requires the two conditions; then, we arrange the likelihood function to exhibit the structure of a Poisson distribution likelihood.

The following Theorem 2 is a restated version of Theorem 1, where we use standard Poisson process notation, and state mathematically the two conditions we introduce in Theorem 1.

Theorem 2. Consider a Poisson process $\{X(t), T \geq t \geq 0\}$ with rate λ (where $X(t)$ denotes the number of jumps up to time t), and denote the jumping times $\{T_1, T_2, \dots\}$. Denote the random variables of the start and end of an observation period S and E , and suppose S and E satisfy the following conditions, respectively:

Condition 1: Given $T_1 = t_1$, S is defined on $[t_1, T]$, and

$$P(S = t | T_1 = t_1) = g(t), \forall T \geq t \geq t_1,$$

where $g(t)$ is not dependent on λ or the sample path of the Poisson process after t_1 .

Condition 2: Given the jumping times of the Poisson process up to any time t , that is, for some $m \in \mathbb{Z}^+$, given $\{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}$, the distribution of E is

$$P(E = t | \{T_1 = t_1, \dots, T_m = t_m, T_{m+1} > t\}) = h_m(t), \text{ for some } m \in \mathbb{Z}^+, \forall T \geq t \geq t_m,$$

where $h_m(t)$ is a function independent of λ . Furthermore, given $T_1 = t_1$, the distribution of E is independent of the realization of S :

$$P(E = t | T_1 = t_1, S = s) = P(E = t | T_1 = t_1), \forall T \geq t \geq t_1, \forall s \geq t_1.$$

Now, suppose we observe data which contains for each incident, $S = s$, $E = e$, $X(e) - X(s) = M$, where $X(s) = m \geq 1$, and all the jumping times within interval $(s, e]$: $JM = \{T_{m+1} = t_{m+1}, \dots, T_{m+M} = t_{m+M}\}$, where $t_{m+1} > s$ and $t_{m+M} \leq e$. Then, conditional on the observed jumping times before s : $H := \{T_1 = t_1, \dots, T_m = t_m \leq s\}$, the likelihood of the data can be decomposed as follows:

$$P(X(e) - X(s) = M, E = e, S = s, JM | H) = f(e, s, H, JM)p(M | \lambda(e - s)), \quad (8)$$

where $p(M | \lambda(e - s))$ denotes the likelihood of a Poisson distribution with M occurrences and rate parameter $\lambda(e - s)$, and $f(e, s, H, JM)$ is a function that does not depend on λ .

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```
∀ (T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count) (D : M.EndpointDensityPresentation)
{rate : ℝ},
0 < rate →
0 < T.window.exposure →
  ∀ {Omega : Type u_1} [inst_1 : MeasurableSpace Omega] [inst_2 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_3 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_2}
  [inst_4 : MeasureTheory.SFinite startReference]
  (selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail)
  (G : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
M.selectedStartLikelihood = G.selectedStartLikelihood →
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D rate =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate T.window.exposure T.count λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.RateIndependent](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

```
1. parameter: AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline
2. parameter: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count
3. parameter: M.EndpointDensityPresentation
4. parameter: ℝ
5. assumption: 0 < rate
6. assumption: 0 < T.window.exposure
7. parameter: Type u_1
8. parameter: MeasurableSpace Omega
9. assumption: StandardBorelSpace Omega
10. parameter: MeasureTheory.Measure Omega
11. assumption: MeasureTheory.IsProbabilityMeasure P
12. parameter: Type u_2
13. parameter: MeasurableSpace Tail
14. parameter: LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail
15. parameter: MeasureTheory.Measure NNReal
16. assumption: MeasureTheory.SFinite startReference
17. parameter: LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T
18. assumption: M.selectedStartLikelihood = G.selectedStartLikelihood
19. conclusion: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D rate =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate T.window.exposure T.count Λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceTheorem1LikelihoodDecomposition

Recorded source-to-Spec screening Verdict: matches

Reason: It gives Equation (8)’s fixed-history likelihood as a rate-independent residual times the Poisson mass at rate times $(e - s)$. The bridge identifies the selected-start factor with a proper conditional density $g(s \mid T_1)$, tied to the timeline start, with Condition-1 support and future-tail independence retained.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 291-323

Verbatim source input

Then, the likelihood of the data D_i as a function of the reporting rate λ , conditional on all reports before s_i , can be decomposed as

$$L(\lambda|D_i) = p$$

◆
~

$$M_i; \lambda(e_i - s_i)$$

◆
~

$$f(D_i),$$

(1)

where

p

◆
~

$$M_i; \lambda(e_i - s_i)$$

◆
~

$$= [\lambda(e_i - s_i)]$$

~

$$M_i$$

~

$$M_i! \exp(-\lambda(e_i - s_i))$$

(2)

5

[form-feed in source extraction]

is the probability mass function of a Poisson distributed random variable with M_i occurrences and rate parameter $\lambda(e_i - s_i)$, and $f(D_i)$ is a function that does not depend on λ .

[Next verbatim source excerpt]

As reference, the likelihood of a Poisson distribution with $M \geq 0$ occurrences and rate $\lambda(e - s) > 0$ is as follows:

$$p(M|\lambda(e - s)) = [\lambda(e - s)]^M$$

M!

$$\exp(-\lambda(e - s)).$$

Expanded PaperInterface specification

$\forall$ (rate exposure : $\mathbb{R}$) (count : $\mathbb{N}$),
0 < rate * exposure →
LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate exposure count =
Real.exp (-(rate * exposure)) * (rate * exposure) ^ count / ↑count.factorial

Retained governing declarations

- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. parameter: $\mathbb{N}$
4. assumption: 0 < rate * exposure
5. conclusion: LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain rate exposure count = Real.exp (-(rate * exposure)) * (rate * exposure) ^ count / ↑count.factorial

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation2PoissonCountPMF

Recorded source-to-Spec screening Verdict: matches

Reason: Equation (2) gives the exponential/factorial Poisson mass for positive lambda times exposure; the Spec is exactly that count-likelihood expansion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 325-334

Verbatim source input

The decomposition in Equation (1) enables standard Poisson rate estimation methods to be applied. For example, the maximum likelihood estimate for λ is:

$$\hat{\lambda}_{MLE} = \frac{\sum_i M_i}{\sum_i (e_i - s_i)}.$$

(3)

Expanded PaperInterface specification

```

∀ (totalCount : ℕ) (totalExposure : ℝ),
  0 < totalExposure →
    LBG24SpatialUnderreporting.mleRate totalCount totalExposure = ↑totalCount / totalExposure ∧
    (totalCount = 0 →
      ∀ (rate : ℝ),
        0 ≤ rate →
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
            LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
              (LBG24SpatialUnderreporting.mleRate totalCount totalExposure)) ∧
    (totalCount ≠ 0 →
      ∀ (rate : ℝ),
        0 < rate →
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
            LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
              (LBG24SpatialUnderreporting.mleRate totalCount totalExposure))

```

Retained governing declarations

- [LBG24SpatialUnderreporting.mleRate](#)
- [LBG24SpatialUnderreporting.poissonRateLogLikelihood](#)

Lean-elaborated claim roles

```

1. parameter: ℕ
2. parameter: ℝ
3. assumption: 0 < totalExposure
4. conclusion: LBG24SpatialUnderreporting.mleRate totalCount totalExposure = ↑totalCount / totalExposure ∧
  (totalCount = 0 →
    ∀ (rate : ℝ),
      0 ≤ rate →
        LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
            (LBG24SpatialUnderreporting.mleRate totalCount totalExposure)) ∧
  (totalCount ≠ 0 →
    ∀ (rate : ℝ),
      0 < rate →
        LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure rate ≤
          LBG24SpatialUnderreporting.poissonRateLogLikelihood totalCount totalExposure
            (LBG24SpatialUnderreporting.mleRate totalCount totalExposure))

```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation3MaximumLikelihoodEstimate

Recorded source-to-Spec screening Verdict: matches

Reason: Equation (3) gives total duplicate count divided by total exposure as the MLE; the Spec states that formula and its Poisson log-likelihood maximum on the relevant rate branches.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 920-944

Verbatim source input

First, we assume that the reporting rate is affected by the type vector θ as in a Poisson regression:
for each model, we have a standard reporting rate $\lambda\theta$, defined as:

$$\lambda\theta = \exp$$

$$\alpha + \beta^T \theta$$

(5)

and so, for each incident i , the likelihood follows from Equation (1) and satisfies the following:

$$L(\lambda\theta_i|D_i) \propto p$$

$$M_i; \lambda\theta_i(e_i - s_i)$$

(6)

13

[form-feed in source extraction]

where β is a coefficient vector of appropriate dimension, (α, β) are coefficients to be estimated, and $p(\cdot)$ is the Poisson likelihood function as defined in Equation (2).

Expanded PaperInterface specification

```
∀ {Feature : Type u_1} [inst : Fintype Feature] (alpha : ℝ) (beta theta : Feature → ℝ)
(T : AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline)
(M : LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count)
(D : M.EndpointDensityPresentation),
0 < T.window.exposure →
∀ {Omega : Type u_2} [inst_1 : MeasurableSpace Omega] [inst_2 : StandardBorelSpace Omega]
{P : MeasureTheory.Measure Omega} [inst_3 : MeasureTheory.IsProbabilityMeasure P] {Tail : Type u_3}
[inst_4 : MeasurableSpace Tail]
(selection : LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail)
{startReference : MeasureTheory.Measure NNReal} [inst_5 : MeasureTheory.SFinite startReference]
(G : LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T),
M.selectedStartLikelihood = G.selectedStartLikelihood →
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D
  (LBG24SpatialUnderreporting.poissonRegressionRate alpha beta theta) =
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
  LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain
  (Real.exp (alpha + ∑ j : Feature, beta j * theta j)) T.window.exposure T.count Λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
  LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ObservationWindow.exposure](#)
- [AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline](#)
- [AppliedModelingLib.Probability.PoissonProcess.RateIndependent](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.EndpointDensityPresentation](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood](#)
- [LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)
- [LBG24SpatialUnderreporting.poissonRegressionRate](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Fintype Feature
3. parameter: ℝ
4. parameter: Feature → ℝ
5. parameter: Feature → ℝ
6. parameter: AppliedModelingLib.Probability.PoissonProcess.OrderedFiniteJumpTimeline
7. parameter: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel T.count
8. parameter: M.EndpointDensityPresentation
9. assumption: 0 < T.window.exposure
10. parameter: Type u_2
11. parameter: MeasurableSpace Omega
12. assumption: StandardBorelSpace Omega
13. parameter: MeasureTheory.Measure NNReal
14. assumption: MeasureTheory.IsProbabilityMeasure P
15. parameter: Type u_3
16. parameter: MeasurableSpace Tail
17. parameter: LBG24SpatialUnderreporting.Theorem2ConditionOneSelection Omega P Tail
18. parameter: MeasureTheory.Measure NNReal
19. assumption: MeasureTheory.SFinite startReference
20. parameter: LBG24SpatialUnderreporting.Theorem2ConditionOneFixedHistoryDensityPresentation selection startReference T
21. assumption: M.selectedStartLikelihood = G.selectedStartLikelihood
22. conclusion: LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.conditionalLikelihood T M D
    (LBG24SpatialUnderreporting.poissonRegressionRate alpha beta theta) =
    LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D *
    LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain (Real.exp (alpha + ∑ j : Feature, beta j * theta j))
    T.window.exposure T.count Λ
AppliedModelingLib.Probability.PoissonProcess.RateIndependent fun (_rate : ℝ) =>
    LBG24SpatialUnderreporting.AppendixTheorem2CausalStoppingSourceModel.rateFreeResidual T M D

```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceEquation6PoissonRegressionLikelihood

Recorded source-to-Spec screening Verdict: matches

Reason: It substitutes Equation (5)'s log link $\exp(\alpha + \beta^T \theta)$ into the Equation (8) decomposition, yielding Equation (6)'s Poisson mass with a rate-independent residual. The explicit equality witnesses the source's proportional-likelihood notation and retains the selected-start density bridge.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 336-339

Verbatim source input

The proof is in Appendix B.2 and follows from decomposing the likelihood function into full conditional probabilities. The key step is to find the conditions under which one can use data-dependent observation period length $E_i - S_i$, while maintaining a Poisson likelihood for reports within the period.⁵ The result extends to the zero-inflated model we use in our empirical estimation.

[Next verbatim source excerpt]

We further fit a zero-inflated Poisson regression [19], which corresponds to a data generating process in which, independently for each incident, only one report can be generated with probability γ , and thus there are 0 duplicate reports:

$M_i = 0.9$

For example, perhaps those incidents are personal requests for service (e.g., to plant a tree somewhere or to prune a tree to enable certain construction), for which duplicates are unlikely; alternatively, officials may physically mark the location of an incident as “report received; full inspection pending,” preventing future reports. Under this case, the likelihood of the data satisfies:

$L(\lambda\theta_i|D_i) \propto$

$(1 - \gamma)^p$

$M_i; \lambda\theta_i(e_i - s_i)$

$\sim$

$M_i \geq 1;$

$\gamma + (1 - \gamma)p(0; \lambda\theta_i(e_i - s_i)) ,$

$M_i = 0.$

(7)

Expanded PaperInterface specification

```
∀ (M : LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel),
  LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure 0 =
    M.gamma + (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure 0 ∧
  (∀ (count : ℕ),
    1 ≤ count →
      LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure count =
        (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure count) ∧
  ∀ {Incident : Type u_1} (incidents : Finset Incident)
    (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ),
    LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood incidents model count =
      ∏ i ∈ incidents,
        LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
        (count i)
```

Retained governing declarations

- [LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel](#)
- [LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain](#)
- [LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood](#)
- [LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood](#)

Lean-elaborated claim roles

```
1. parameter: LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel
2. conclusion: LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure 0 =
  M.gamma + (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure 0 ∧
  (∀ (count : ℕ),
    1 ≤ count →
      LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood M.gamma M.rate M.exposure count =
        (1 - M.gamma) * LBG24SpatialUnderreporting.sourcePoissonPMFOnSourceDomain M.rate M.exposure count) ∧
  ∀ {Incident : Type u_1} (incidents : Finset Incident)
    (model : Incident → LBG24SpatialUnderreporting.ZeroInflatedIncidentSourceModel) (count : Incident → ℕ),
    LBG24SpatialUnderreporting.zeroInflatedIndependentIncidentFamilyLikelihood incidents model count =
      ∏ i ∈ incidents,
        LBG24SpatialUnderreporting.zeroInflatedIncidentLikelihood (model i).gamma (model i).rate (model i).exposure
        (count i)
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceZeroInflatedLikelihoodExtension

Recorded source-to-Spec screening Verdict: matches

Reason: Equation (7)'s zero and positive count mixture branches and independent incident product are exactly the branches and product in the Spec.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 1660-1770

Verbatim source input

Lemma 1. Suppose each incident gets reported independently, and the distribution of the interval of reporting duration T_i has density function $f(\cdot) : [0, \infty) \rightarrow \mathbb{R}$. Then under steady state, $N_{\text{observed}} \theta$ follows a Poisson process with rate Λ' , where

$$\Lambda' = \Lambda \theta$$

$$\int_0^\infty$$

$$\exp$$

$$- \int_0^t$$

$$\int_0^\infty \lambda \theta(u) du$$

$$\int_0^\infty f(t) dt$$

$$\cdot$$

In the simplest time-homogeneous case, this rate simplifies to:

$$\Lambda' = \Lambda \theta$$

$$\int_0^\infty$$

$$\exp(-\lambda \theta t) f(t) dt$$

$$\cdot$$

Proof of Lemma 1. Let $m(t)$ be the number of times an incident is reported in an interval of t , starting from its birth. We know from the model assumption that $m(t)$ follows a Poisson distribution:

$$m(t) \sim \text{Poisson}(\Lambda \theta t)$$

$$\int_0^\infty \lambda \theta(u) du$$

$$\cdot$$

23

[form-feed in source extraction]

In steady state, each unique incident gets reported with probability

$$p = \int_0^\infty$$

$$\Pr[m(t) \geq 1 | T_i = t] f(t) dt$$

$$= \int_0^\infty$$

$$0$$

$$1 - \exp$$

$$- \int_0^t$$

$$\int_0^\infty \lambda \theta(u) du$$

$$\int_0^\infty f(t) dt$$

$$= 1 - \int_0^\infty$$

$$\exp$$

$$- \int_0^t$$

$\int_0^t \lambda \theta(u) du$
 $\int_0^t f(t) dt,$

which simplifies under time-homogeneity to

$p = 1 -$
 $\int_0^\infty$

$\int_0^t \exp(-\lambda \theta t) f(t) dt.$

Over a time interval of t , the total number of incidents of type θ that happen, $N_\theta(t)$ follows a Poisson process with rate $\Lambda \theta$.

Conditional on $N_\theta(t) = n$, $n = 0, 1, \dots$, under steady state, N_{observed}

θ
 (t) follows a binomial distribution with parameters (n, p) .
Thus N_{observed}

θ
 (t) follows a Poisson distribution with rate $\Lambda \theta p$, which completes the proof.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```

∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] {P : MeasureTheory.Measure Omega}
(M : LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Omega P),
(∀ (t : NNReal), Measurable (M.calendarFirstReports.calendarCount t)) ∧
(∀m (omega : Omega) ∂P, M.calendarFirstReports.calendarCount 0 omega = 0) ∧
(∀m (omega : Omega) ∂P, Monotone fun (t : NNReal) => M.calendarFirstReports.calendarCount t omega) ∧
ProbabilityTheory.HasIndepIncements M.calendarFirstReports.calendarCount P ∧
∀ {s t : NNReal},
s ≤ t →
  ∀ (observedCount : ℕ),
P.real
  {omega : Omega |
    M.calendarFirstReports.calendarCount t omega - M.calendarFirstReports.calendarCount s omega =
      observedCount} =
AppliedModelingLib.Probability.PoissonProcess.countLikelihood
(LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity
  M.calendarFirstReports.incidentRate (fun (u : ℝ) => ∫ (v : ℝ) in 0..u, M.reportingIntensity v)
  M.durationDensity)
(↑t - ↑s) observedCount

```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.countLikelihood](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel](#)
- [LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: MeasurableSpace Omega
3. parameter: MeasureTheory.Measure Omega
4. parameter: LBG24SpatialUnderreporting.Lemma1CalendarTimeDurationSourceModel Omega P
5. conclusion: (∀ (t : NNReal), Measurable (M.calendarFirstReports.calendarCount t)) ∧
(∀m (omega : Omega) ∂P, M.calendarFirstReports.calendarCount 0 omega = 0) ∧
(∀m (omega : Omega) ∂P, Monotone fun (t : NNReal) => M.calendarFirstReports.calendarCount t omega) ∧
ProbabilityTheory.HasIndepIncements M.calendarFirstReports.calendarCount P ∧
∀ {s t : NNReal},
s ≤ t →
  ∀ (observedCount : ℕ),
P.real
  {omega : Omega |
    M.calendarFirstReports.calendarCount t omega - M.calendarFirstReports.calendarCount s omega =
      observedCount} =
AppliedModelingLib.Probability.PoissonProcess.countLikelihood
(LBG24SpatialUnderreporting.continuousDurationObservedIncidentRateOfCumulativeIntensity
  M.calendarFirstReports.incidentRate (fun (u : ℝ) => ∫ (v : ℝ) in 0..u, M.reportingIntensity v)
  M.durationDensity)
(↑t - ↑s) observedCount

```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceLemma1CalendarTimeDurationObservedProcess

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 1 gives the same duration-mixture observed rate and Poisson process under the queue's embedded calendar-time convention.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 1626-1658

Verbatim source input

Proposition 1. Consider the simplest setting: both incident occurrence and reporting follow time-homogeneous Poisson processes, with $\Lambda\theta(\tau) = \Lambda\theta$ and $\lambda\theta(\tau) = \lambda\theta$. Further, let incident reporting duration T_i be distributed according to F . Then using just the number of observed incidents

$N_{\text{observed}}^\theta$
(T) of type θ in a known time duration $[0, T]$, the reporting rate $\lambda\theta$ is not identifiable. In other words, $N_{\text{observed}}^\theta$
(T) is a function of both $\Lambda\theta$ and $\lambda\theta$:

$$\lim_{T \rightarrow \infty} \frac{N_{\text{observed}}^\theta(T)}{T} = \Lambda'_{\theta}$$

where
 $\Lambda'_{\theta} = \Lambda\theta$

$$\frac{1}{\infty} \int_0^\infty \exp(-\lambda\theta t) dF(t)$$

The proof of Proposition 1 follows directly from the following Lemma.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Additional assumptions The Proposition 1 nonidentifiability construction is checked on the nondegenerate observed-count-rate branch $0 < \text{observedRate}$.

Expanded PaperInterface specification

```
∀ {Omega1 : Type u_1} {Omega2 : Type u_2} [inst : MeasurableSpace Omega1] [inst_1 : MeasurableSpace Omega2]
{P1 : MeasureTheory.Measure Omega1} {P2 : MeasureTheory.Measure Omega2}
(M1 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega1 P1)
(M2 : LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega2 P2) (observedRate : ℝ),
0 < observedRate →
  M1.reportingRate ≠ M2.reportingRate →
  M1.durationDensity = M2.durationDensity →
  0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity →
  0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity →
  M1.calendarFirstReports.incidentRate =
    observedRate /
    LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate
    M1.durationDensity →
  M2.calendarFirstReports.incidentRate =
    observedRate /
    LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate
    M2.durationDensity →
  (∀m (omega : Omega1) ∂P1,
    Filter.Tendsto (fun (n : ℕ) => ↑(M1.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop
    (nhds observedRate)) ∧
  (∀m (omega : Omega2) ∂P2,
    Filter.Tendsto (fun (n : ℕ) => ↑(M2.observedUniqueIncidentCount n omega) / ↑n) Filter.atTop
    (nhds observedRate)) ∧
  M1.reportingRate ≠ M2.reportingRate ∧
  ∃ (observedProcess1 :
    AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw
      Omega1 P1)
  (observedProcess2 :
    AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw
      Omega2 P2),
  observedProcess1.count = M1.calendarFirstReports.calendarCount ∧
  observedProcess2.count = M2.calendarFirstReports.calendarCount ∧
  observedProcess1.rate = observedRate ∧ observedProcess2.rate = observedRate
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw](#)
- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel](#)

- [LBG24SpatialUnderreporting.CalendarFirstReportSourceModel.calendarCount](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel](#)
- [LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel.observedUniqueIncidentCount](#)
- [LBG24SpatialUnderreporting.continuousDurationFirstReportProbability](#)

Lean-elaborated claim roles

```

1. parameter: Type u_1
2. parameter: Type u_2
3. parameter: MeasurableSpace Omega1
4. parameter: MeasurableSpace Omega2
5. parameter: MeasureTheory.Measure Omega1
6. parameter: MeasureTheory.Measure Omega2
7. parameter: LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega1 P1
8. parameter: LBG24SpatialUnderreporting.Proposition1CalendarTimeDurationSourceModel Omega2 P2
9. parameter: ℝ
10. assumption: 0 < observedRate
11. assumption: M1.reportingRate ≠ M2.reportingRate
12. assumption: M1.durationDensity = M2.durationDensity
13. assumption: 0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity
14. assumption: 0 < LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity
15. assumption: M1.calendarFirstReports.incidentRate =
  observedRate / LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M1.reportingRate M1.durationDensity
16. assumption: M2.calendarFirstReports.incidentRate =
  observedRate / LBG24SpatialUnderreporting.continuousDurationFirstReportProbability M2.reportingRate M2.durationDensity
17. conclusion: (∀m (omega : Omega1) ∂P1,
  Filter.Tendsto (fun (n : ℕ) => ↑ (M1.observedUniqueIncidentCount n omega) / ↑ n) Filter.atTop (nhds observedRate)) ∧
(∀m (omega : Omega2) ∂P2,
  Filter.Tendsto (fun (n : ℕ) => ↑ (M2.observedUniqueIncidentCount n omega) / ↑ n) Filter.atTop (nhds observedRate)) ∧
M1.reportingRate ≠ M2.reportingRate ∧
∃ (observedProcess1 :
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Omega1 P1)
(observedProcess2 :
  AppliedModelingLib.Probability.PoissonProcess.ForwardHomogeneousPoissonCountingProcessByLaw Omega2 P2),
observedProcess1.count = M1.calendarFirstReports.calendarCount ∧
observedProcess2.count = M2.calendarFirstReports.calendarCount ∧
observedProcess1.rate = observedRate ∧ observedProcess2.rate = observedRate

```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceProposition1CalendarTimeNonidentifiability

Recorded source-to-Spec screening Verdict: matches

Reason: Proposition 1 pairs distinct reporting rates with offsetting incidence rates to give the same observed Poisson rate and long-run count rate, exactly as the Spec encodes.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 1870-1875

Verbatim source input

The proof of the theorem relies on the following lemma, which establishes the distribution of the (random) time between S and the first jump time after S .

Lemma 2. If the distribution of S satisfies condition 1, then the distribution of the time between any realization of S and the next jump time $TX(S)+1$ is an exponential random variable with rate λ .

[Next verbatim source excerpt]

differentially report those problems to the government. Such differences might emerge, for example, due to access to communications technologies, familiarity with the system, or trust in government [3]. As discussed above, the biggest challenge to studying under-reporting is that researchers rarely

[Next verbatim source excerpt]

Condition 1: Given $T1 = t1$, S is defined on $[t1, T]$, and

$P(S = t|T1 = t1) = g(t)$, $\forall T \geq t \geq t1$,

where $g(t)$ is not dependent on λ or the sample path of the Poisson process after $t1$.

Settled source reading The result-specific conditions and corrections are stated in the [clarification memo](docs/SOURCE_CLARIFICATIONS.md).

Expanded PaperInterface specification

```
∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [inst_1 : StandardBorelSpace Omega]
  {P : MeasureTheory.Measure Omega} [inst_2 : MeasureTheory.IsProbabilityMeasure P]
  (M : LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Omega P) (u : NNReal),
  ∀m (omega : Omega) ∂P,
  (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (M.interarrivalPath omega)).toNNReal ≤
  M.observationHorizon →
  ((ProbabilityTheory.condExpKernel P (MeasurableSpace.comap M.selection.firstReportTime inferInstance)) omega).real
  {omega' : Omega |
    AppliedModelingLib.Probability.PoissonProcess.forwardPostStopIntervalCount M.process M.selection.startTime u
    omega' =
    0} =
  ({ rate := M.rate, rate_pos := M.rate_pos }.measure (Set.Ioi ↑u)).toReal
```

Retained governing declarations

- [AppliedModelingLib.Probability.Exponential.Model](#)
- [AppliedModelingLib.Probability.Exponential.Model.measure](#)
- [AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.forwardPostStopIntervalCount](#)
- [LBG24SpatialUnderreporting.Lemma2ForwardSourceModel](#)
- [LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.process](#)
- [LBG24SpatialUnderreporting.Lemma2ForwardSourceModel.selection](#)
- [LBG24SpatialUnderreporting.Theorem2ConditionOneSelection](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: MeasurableSpace Omega
3. assumption: StandardBorelSpace Omega
4. parameter: MeasureTheory.Measure Omega
5. assumption: MeasureTheory.IsProbabilityMeasure P
6. parameter: LBG24SpatialUnderreporting.Lemma2ForwardSourceModel Omega P
7. parameter: NNReal
8. conclusion: ∀m (omega : Omega) ∂P,
  (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (M.interarrivalPath omega)).toNNReal ≤
  M.observationHorizon →
  ((ProbabilityTheory.condExpKernel P (MeasurableSpace.comap M.selection.firstReportTime inferInstance)) omega).real
  {omega' : Omega |
    AppliedModelingLib.Probability.PoissonProcess.forwardPostStopIntervalCount M.process M.selection.startTime u
    omega' =
    0} =
  ({ rate := M.rate, rate_pos := M.rate_pos }.measure (Set.Ioi ↑u)).toReal
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourceLemma2SelectedStartExponentialTail

Recorded source-to-Spec screening Verdict: matches

Reason: Lemma 2's next-report wait after a Condition-1 selected start is the equivalent rate-exponential conditional no-arrival tail stated by the Spec.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 4402-4420

Verbatim source input

In Section 6.2 and in Appendix D.8.1, the predictions were generated using the results of the spatial model we presented in the main text. The training set used to obtain such results was constructed by defining the time of the first report as the start of our observation period, as detailed in Appendix D.2. However, our theoretical analysis also holds if we instead use as the Poisson interval the time starting after the second report – if we split a Poisson process on the time of the first jump, and start counting subsequent jumps at that time, the resulting counting process would still be a Poisson process with the same rate. Thus, we can estimate the Poisson process rate using our methods, starting with the second report time.²⁵ However, crucially, we as researchers know for the incident the time between the first and second report, which was not used to train the model (but which the model is trying to predict). Intuitively, this procedure evaluates our methods in a setting in which the ground truth is known.

²⁵Formally, consider a Poisson process with rate λ , denoted as $\{X(t), t \geq 0\}$, with the first jump time denoted as T_1 . Based on the memoryless property of Poisson process jump times, the process $\{X(t) - 1, t \geq T_1\}$ is still a Poisson process with the same rate parameter (see e.g., [34]). Thus in our context, for an incident with type θ , if we treat the first report as the start of a Poisson process, treat the second report as equivalent to the first report, and count duplicates starting at the third report, this reporting process is still a Poisson process with rate $\lambda\theta$, but now the start of this process is observed.

Expanded PaperInterface specification

```
∀ {Omega : Type u_1} [inst : MeasurableSpace Omega] [StandardBorelSpace Omega] {P : MeasureTheory.Measure Omega}
[inst_2 : MeasureTheory.IsProbabilityMeasure P] {rate : ℝ},
0 < rate →
  ∀ (interarrivalPath : Omega → ℕ → ℝ),
  Measurable interarrivalPath →
  ProbabilityTheory.HasLaw interarrivalPath
  (AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate) P →
  ↑ (ProbabilityTheory.condDistrib
    (fun (omega : Omega) =>
      AppliedModelingLib.Probability.PoissonProcess.futureInterarrival 1 (interarrivalPath omega))
    (fun (omega : Omega) =>
      (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival
        (interarrivalPath omega)).toNNReal)
    P) == "[MeasureTheory.Measure.map
  (fun (omega : Omega) =>
    (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (interarrivalPath omega)).toNNReal)
  P]
  ↑ (ProbabilityTheory.Kernel.const NNReal
    (AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate))
```

Retained governing declarations

- [AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival](#)
- [AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure](#)
- [AppliedModelingLib.Probability.PoissonProcess.futureInterarrival](#)

Lean-elaborated claim roles

```
1. parameter: Type u_1
2. parameter: MeasurableSpace Omega
3. assumption: StandardBorelSpace Omega
4. parameter: MeasureTheory.Measure Omega
5. assumption: MeasureTheory.IsProbabilityMeasure P
6. parameter: ℝ
7. assumption: 0 < rate
8. parameter: Omega → ℕ → ℝ
9. assumption: Measurable interarrivalPath
10. assumption: ProbabilityTheory.HasLaw interarrivalPath
    (AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate) P
11. conclusion: ↑ (ProbabilityTheory.condDistrib
  (fun (omega : Omega) =>
    AppliedModelingLib.Probability.PoissonProcess.futureInterarrival 1 (interarrivalPath omega))
  (fun (omega : Omega) =>
    (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (interarrivalPath omega)).toNNReal)
  P) == "[MeasureTheory.Measure.map
  (fun (omega : Omega) =>
    (AppliedModelingLib.Probability.PoissonProcess.canonicalFirstArrival (interarrivalPath omega)).toNNReal)
  P]
  ↑ (ProbabilityTheory.Kernel.const NNReal
    (AppliedModelingLib.Probability.PoissonProcess.exponentialInterarrivalMeasure rate))
```

Lean proof endpoint (built separately)

LBG24SpatialUnderreporting.sourcePostFirstJumpPoissonShift

Recorded source-to-Spec screening Verdict: matches

Reason: Appendix D.8.2 says the process shifted after the first jump remains same-rate Poisson; the generic HasLaw premise gives the equivalent unchanged future iid-exponential-sequence conditional law.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation